ANNEX 1 (part A)

Coordination and support action

NUMBER — 767412 — SPRING

Table of Contents

1.1. The project summary....3
1.2. The list of beneficiaries....4
1.3. Workplan Tables - Detailed implementation....5
1.3.1. WT1 List of work packages....5
1.3.2. WT2 List of deliverables....6
1.3.3. WT3 Work package descriptions....10
Work package 1.... 10
Work package 2.... 13
Work package 3.... 17
Work package 4.... 20

Work package 5 24
Work package 6 27
1.3.4. WT4 List of milestones 28
1.3.5. WT5 Critical Implementation risks and mitigation actions 29
1.3.6 WT6 Summary of project effort in person-months 30
1.3.7. WT7 Tentative schedule of project reviews 31

1.1. The project summary

Project Number [1]	767412	Project Acronym [2]	SPRING

One form per project

General information

Project title [3]	Setting the framework for the enhanced impact of SPIRE projects
Starting date [4]	01/09/2017
Duration in months [5]	24
Call (part) identifier [6]	H2020-SPIRE-2017
Topic	SPIRE-11-2017 Support for the enhancement of the impact of SPIRE PPP projects
Fixed EC Keywords	Financial Engineering
Free keywords	Enhancing impact of SPIRE projects
Abstract [7]	

Increasing industrial uptake of project findings is at the heart of project SPRING. It is the essential building block for ensuring greater impact of SPIRE projects and therefore progress towards the SPIRE roadmap goals of increased resource and energy efficiency in the EU process industries.

Instead of focusing on a small cluster of projects, SPRING has been developed to provide the mechanism to enhance the impact of all SPIRE projects.

Project SPRING's objective is to increase progress towards the SPIRE goals and enhance return on investment in projects by addressing the needs of those who make the decisions to adopt process innovations in industry and barriers to their adoption. It will do this by providing guidance to project participants, decision makers in industry and broader SPIRE stakeholders, enabling them collectively to:

1. Improve the articulation of the value of project exploitable outputs
2. Improve the articulation of industry needs and barriers-to-uptake of exploitable outputs
3. Improve the mapping of project value to industry needs
4. Identify policy gaps and recommendations to improve project impact

To address these objectives, the project will deliver six sets of exploitable outputs:

A – Guidance on best practice of how to measure progress, impact and success of SPIRE projects.

B – Frameworks for getting different levels of project results to the right audience through the spire2030.eu portalC – A model for mapping project outputs to industry needs, through thematic, interactive industry workshops, expert input and technology scanning methods.

D – Guidance for understanding business barriers to uptake, including best practice for enabling good decision makingwhen evaluating project outputs.

E – A package of training and network groups to upskill SPIRE project participantsF – Identification of policy gaps and future SPIRE needs

1.2. List of Beneficiaries

Project Number [1]	767412	Project Acronym [2]	SPRING

List of Beneficiaries					
No	**Name**	**Short name**	**Country**	**Project entry month[8]**	**Project exit month**
1	BRITEST LBG	BRITEST	United Kingdom	1	24
2	INEOS EUROPE AG	INEOS	Switzerland	1	24
3	A.SPIRE	ASPIRE	Belgium	1	24
4	INNOVACIO I RECERCA INDUSTRIAL I SOSTENIBLE SL	IRIS	Spain	1	24
5	UNIVERSITY OF LEEDS	LEEDS	United Kingdom	1	24
6	Centro di Ricerca e Innovazione tecnologica srl	CRIT	Italy	1	24

1.3. Workplan Tables - Detailed implementation

1.3.1. WT1 List of work packages

WP Number[9]	WP Title	Lead beneficiary[10]	Personmonths[11]	Start month[12]	End month[13]
WP1	Coordination and Management	1 - BRITEST		1	24
WP2	Research	5 - LEEDS		1	13
WP3	Testing and Validation	4 - IRIS		1	21
WP4	Output Generation	1 - BRITEST		4	24
WP5	Stakeholder Engagement	3 - ASPIRE		1	24
WP6	Ethics requirements	1 - BRITEST		1	24
		Total			

1.3.2. WT2 list of deliverables

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D1.1	Secure project portal, website and extranet established	WP1	1 - BRITEST	Websites, patents filling, etc.	Public	1
D1.2	Draft Data Management Plan	WP1	1 - BRITEST	Report	Confidential, only for members of the consortium (including the Commission Services)	6
D1.3	Draft Exploitation Plan	WP1	1 - BRITEST	Report	Confidential, only for members of the consortium (including the Commission Services)	12
D1.4	Year 1 Management Report and Financing Plan	WP1	1 - BRITEST	Report	Confidential, only for members of the consortium (including the Commission Services)	12
D1.5	Final Data Management Plan	WP1	1 - BRITEST	Report	Public	24
D1.6	Final Exploitation Plan	WP1	1 - BRITEST	Report	Public	24
D1.7	Establish Dissemination Plan and Activities Tracker	WP1	1 - BRITEST	Report	Confidential, only for members of the consortium (including the Commission Services)	3
D1.8	Open Research Data Pilot	WP1	1 - BRITEST	ORDP: Open Research Data Pilot	Confidential, only for members of the consortium (including the Commission Services)	6
D2.1	Review of metrics and other good practice for measuring project progress	WP2	5 - LEEDS	Report	Confidential, only for members of the consortium (including the Commission Services)	3

D2.2	Initial review of highlevel SPIRE and case study project outputs	WP2	1 - BRITEST	Report	Confidential, only for members of the consortium (including the Commission Services)	3

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D2.3	Review of approaches to structuring information for educational purposes	WP2	5 - LEEDS	Report	Confidential, only for members of the consortium (including the Commission Services)	3
D2.4	Review of available technology scanning approaches	WP2	6 - CRIT	Report	Confidential, only for members of the consortium (including the Commission Services)	6
D2.5	Review of best practice decision making on project outputs	WP2	5 - LEEDS	Report	Confidential, only for members of the consortium (including the Commission Services)	7
D2.6	Review of barriers to industrial implementation	WP2	1 - BRITEST	Report	Confidential, only for members of the consortium (including the Commission Services)	8
D2.7	Review of high-level outputs for SPIRE projects	WP2	1 - BRITEST	Report	Confidential, only for members of the consortium (including the Commission Services)	13
D2.8	Draft methodology for impact assessment and enhancement	WP2	5 - LEEDS	Report	Confidential, only for members of the consortium (including the Commission Services)	13

D3.1	Case studies testing report 1	WP3	4 - IRIS	Report	Confidential, only for members of the consortium (including the Commission Services)	10
D3.2	Case Studies Testing - Final Report	WP3	4 - IRIS	Report	Confidential, only for members of the consortium (including the Commission Services)	21

Deliverable Number[14]	**Deliverable Title**	**WP number[9]**	**Lead beneficiary**	**Type[15]**	**Dissemination level[16]**	**Due Date (in months)[17]**
D3.3	Summary report of case study project outputs suitable for educational/ training resources	WP3	1 - BRITEST	Report	Confidential, only for members of the consortium (including the Commission Services)	12
D4.1	Structure proposal for capturing high-level findings from SPIRE projects	WP4	1 - BRITEST	Report	Confidential, only for members of the consortium (including the Commission Services)	7
D4.2	Preliminary guidance for best practice measurement of progress/ impact/ success of projects	WP4	1 - BRITEST	Report	Confidential, only for members of the consortium (including the Commission Services)	13
D4.3	Framework proposal for capturing education/ training resources	WP4	1 - BRITEST	Report	Public	12
D4.4	Preliminary structure for thematic workshops	WP4	1 - BRITEST	Report	Confidential, only for members of the consortium (including the Commission Services)	16

D4.5	Preliminary recommendations and gaps report	WP4	1 - BRITEST	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D4.6	Guidance for framing SPIRE projects outputs on spire2030.eu	WP4	1 - BRITEST	Report	Public	20
D4.7	Thematic workshop structure guidance	WP4	1 - BRITEST	Report	Public	21
D4.8	Guidance for best practice measurement of progress, impact and success of SPIRE projects	WP4	1 - BRITEST	Report	Confidential, only for members of the consortium (including the Commission Services)	23
D4.9	Training and presentation package for communicating value and impact of SPIRE projects	WP4	1 - BRITEST	Websites, patents filling, etc.	Public	24

Deliverable Number[14]	**Deliverable Title**	**WP number[9]**	**Lead beneficiary**	**Type[15]**	**Dissemination level[16]**	**Due Date (in months)[17]**
D4.10	Final recommondations and gaps report	WP4	1 - BRITEST	Report	Public	24
D5.1	Guidance for project progress, impact and success measurement published	WP5	3 - ASPIRE	Report	Public	24
D5.2	Thematic Workshops	WP5	1 - BRITEST	Other	Public	22
D5.3	spire2030.eu update to incorporate project output frameworks	WP5	3 - ASPIRE	Websites, patents filling, etc.	Public	24
D5.4	Training package on website	WP5	3 - ASPIRE	Websites, patents filling, etc.	Public	24
D6.1	NEC - Requirement No. 1	WP6	1 - BRITEST	Ethics	Confidential, only for members of the consortium (including the Commission Services)	2

D6.2	POPD - Requirement No. 2	WP6	1 - BRITEST	Ethics	Confidential, only for members of the consortium (including the Commission Services)	2

1.3.3. WT3 Work package descriptions

Work package number [9]	WP1	**Lead beneficiary** [10]	1 - BRITEST
Work package title	Coordination and Management		
Start month	1	**End month**	24

Objectives

The objective of WP1 is to build and maintain coherent and efficient project management over the entire project period. Specific objectives are to:
i) Coordinate and conduct the project according to the work plan, ensure that objectives and timelines are met andbudgets are honoured.
ii) Manage intra-consortium communication - Foster and boost inter-relations between project partners, exchange ofknowledge and information between teams involved in the different WPs and tasks. iii) Ensure a timely and efficient transfer of information (e.g. reports) to and from the Commission.
iv) Distribute the financial contribution paid by the Commission among the partners. Furthermore, this WP includes constant monitoring and adaptation of the project management procedures to improve its effectiveness.

Description of work and role of partners

WP1 - Coordination and Management [Months: 1-24]
BRITEST, INEOS, ASPIRE, IRIS, LEEDS, CRIT
This work package includes the following activities (BRITEST lead, spread across the full 24 months):
- Supervise and monitor the progress of the project activities. Confirm project deliverables and milestones.
- Organise, administer and act as secretariat for project meetings (e.g. General Assembly)
- Maintain the communication and exchange with the European Commission. Management of knowledge generated bythe project in accordance to the provisions included in the EU contract and the Consortium Agreement and coordination of the innovation-related activities.
- Receive the entire financial contribution from the Commission and ensure timely budget allocation to the Contractorsconsistent with the work programme and the decisions taken by the appropriate Bodies. Keep accurate records identifying the budget share allocated to each Contractor and inform the EC of the distribution of funds and the date of transfer to the partners in line with review periods.
- Overall legal and contractual management, including the maintenance of the Consortium Agreement. Organisationand follow-up of major meetings among the partners.
- Organisation of the activity reports to be delivered to the EC, including the collection of contributions from the partners.Submission to the Commission of all reports, project deliverables and cost statements prepared and duly certified by the Contractors.
- Establishment and management of communication procedures and tools for facilitating exchange among partners. Maintenance of contacts with the partners. Tracking of official communications.
- Support of the partners with reference to procedures requested by the European Commission.

Task 1.1: Intra-consortium information and communication sharing (BRITEST lead, M1)
BRITEST will establish and maintain a secure project 'portal' to facilitate the effective communication and distribution of information, reporting and data storage to and amongst project partners. Accessible to project partners only via registration and log-in, this will act as the principal repository for project documentation and provide a platform for work package leaders to communicate and co-ordinate information. It will also be used to facilitate dialogue amongst researchers and industrial partners in the project, information exchange across work packages and promote dissemination of project deliverables via symposia, scientific workshops and conferences.

Task 1.2: Exploitation Plan (BRITEST lead, All Partners, M1-24)

BRITEST will work with project partners to maintain a project exploitation plan and ensure that all partners are effectively supported in understanding EC expectations and the relevant processes/ mechanisms available for exploiting project outputs. The development of exploitable outputs is covered in WP4.

Task 1.3: Data and Knowledge Management (BRITEST lead, All Partners, M3-24)
BRITEST will coordinate the development and maintenance of a project Data Management Plan to make data from the project findable, accessible, interoperable and reusable (FAIR).
BRITEST will work with project partners on a confidential basis to establish a register of background/ foreground IP and an effective knowledge management system to capture all dissemination activities emerging from the project.

Task 1.4: Management of project progress and use of resources (BRITEST lead, M1-M24)
BRITEST will coordinate the management of project progress and use of resources through monthly Work Package Leaders conference calls and collation of resource and finance summaries for project reviews.

Participation per Partner

Partner number and short name	WP1 effort
1 - BRITEST	
2 - INEOS	
3 - ASPIRE	
4 - IRIS	
5 - LEEDS	
6 - CRIT	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D1.1	Secure project portal, website and extranet established	1 - BRITEST	Websites, patents filling, etc.	Public	1
D1.2	Draft Data Management Plan	1 - BRITEST	Report	Confidential, only for members of the consortium (including the Commission Services)	6
D1.3	Draft Exploitation Plan	1 - BRITEST	Report	Confidential, only for members of the consortium (including the Commission Services)	12

D1.4	Year 1 Management Report and Financing Plan	1 - BRITEST	Report	Confidential, only for members of the consortium (including the Commission Services)	12
D1.5	Final Data Management Plan	1 - BRITEST	Report	Public	24
D1.6	Final Exploitation Plan	1 - BRITEST	Report	Public	24
D1.7	Establish Dissemination Plan and Activities Tracker	1 - BRITEST	Report	Confidential, only for members of the consortium (including the Commission Services)	3

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D1.8	Open Research Data Pilot	1 - BRITEST	ORDP: Open Research Data Pilot	Confidential, only for members of the consortium (including the Commission Services)	6

Description of deliverables

D1.1 – Secure project 'portal', website and extranet established (BRITEST – M1)
D1.2 – Draft Data Management Plan (BRITEST, LEEDS – M6)
D1.3 – Draft Exploitation Plan (BRITEST, All Partners – M12)
D1.4 – Year 1 Management Report and Financing Plan – (BRITEST – M12)
D1.5 – Final Data Management Plan (BRITEST, LEEDS – M24)
D1.6 – Final Exploitation Plan (BRITEST, All Partners – M24)
D1.7 – Establish Dissemination Plan and Activities Tracker (BRITEST – M3)

D1.1 : Secure project portal, website and extranet established [1]

Establish the means for internal consortium communication and sharing of data. Establish public facing website for the project (through the spire2030.eu portal)

D1.2 : Draft Data Management Plan [6]

In line with proposal guidelines, a draft Data Management Plan will be developed for management of data arising from research and case study projects.

D1.3 : Draft Exploitation Plan [12]

Draft report providing overview of expected outputs and exploitation from SPRING

D1.4 : Year 1 Management Report and Financing Plan [12]

Report detailing resource and finances used by partners in M1-M12, for internal tracking of overall project progress and resources.

D1.5 : Final Data Management Plan [24]

Summary report detailing final data management plan for information collected in SPRING.

D1.6 : Final Exploitation Plan [24]

Final summary report detailing exploitable outputs from the project.

D1.7 : Establish Dissemination Plan and Activities Tracker [3]

Set up system for planning and tracking dissemination activities for the project.

D1.8 : Open Research Data Pilot [6]

Draft plan to ensure SPRING data meets the needs of the Open Research Data Pilot. Linked to Data Management Plan (D1.2) and finalised in D1.5

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification

Work package number [9]	WP2	**Lead beneficiary** [10]	5 - LEEDS
Work package title	Research		
Start month	1	**End month**	13

Objectives

The objectives of WP2 are to develop a best practice approach to assessing the impact of SPIRE and similar projects which will in turn be integrated with a proactive set of tools and guidance aimed at enhancing project impact applicable to each of the six scales of potential stakeholder interest set out in Section 1.1. Specifically, WP2 will:

i) Develop a framework for impact assessment appropriate to projects of the type undertaken in SPIRE.

ii) Seek to understand current barriers to full implementation of project outputs in industry.

iii) Develop guidance and tools to facilitate effective promotion of project outputs to industrial and other stakeholdersiv) Identify and develop educational resources to facilitate knowledge transfer and training about outputs from SPIRE and similar projects.

v) Integrate the above into a draft methodology for impact assessment and enhancement.

Description of work and role of partners

WP2 - Research [Months: 1-13]

LEEDS, BRITEST, INEOS, ASPIRE, IRIS, CRIT

Task 2.A1: Review metrics for measuring project progress, impact and success from FoF-Impact, SPIRE 4, industry & projects within SPRING (BRITEST, All, M1-2)

BRITEST will lead the work to collate progress metrics and methods from consortium projects to help assess which should be tested on the SPRING case study projects (e.g. proposals from STYLE regarding use of qualitative sustainability metrics at review points).

Task 2.A2: Research other good practice for project progress/ impact/ success metrics (LEEDS, with input from CRIT, M1-3)

LEEDS will lead the work to review good practice in evidencing the progress of research and innovation projects towards impact. It will assess existing approaches in order to synthesise a framework and a set of metrics appropriate to SPIRE projects but embedded within a broad understanding of the innovation process. The framework will support assessment of achieved impact and identification of opportunities for further dissemination and impact-oriented actions. In this latter mode, it would act as a diagnostic tool and prescriptive device, potentially capable of being applied more than once during a project's development path.

Task 2.B1: Review high-level project outputs from case studies and select SPIRE projects (BRITEST, All, M1-3) BRITEST will lead this task to catalogue high level outputs from the case study projects, to enable a structure to be developed to categorise the outputs (in WP4).

Task 2.B4: Review high-level project outputs from wider set of SPIRE projects (BRITEST, ASPIRE, M7-M12) BRITEST will work with ASPIRE to identify suitable additional SPIRE project to have their high-level outputs catalogued, to allow further testing of the structure to categorise outputs (in WP4).

Task 2.B5: Review Educate to Innovate and educational structure resources (LEEDS, with input from BRITEST, M1-3) LEEDS will work with BRITEST to review approaches to structuring information from projects for educational resources. Among others, the Educate to Innovate programme will be reviewed to establish which learning is transferable to general SPIRE projects.

Task 2.C1: Review available technology scanning approaches (CRIT, with input from LEEDS, M3-6)
CRIT will work with LEEDS to review approaches to technology scanning. These are techniques used to ensure that projects are aware of developments in innovations and markets relevant to their subject area, so that projects are not being run in isolation of external developments. Examples include social media threads used in EPOS and commercial tools used by CRIT.

Task 2.D1: Review best practice for enabling good decision making when evaluating project outputs (LEEDS, with input from All, M4-7)
LEEDS will lead this task to support well-founded decisions about possible uptake of project outputs, tailored to each of the six levels indicated in section 1.1. It will incorporate both demand-side activities to scan and establish interest from stakeholders external to the project and supply-side advice about ways for projects to move forward with promoting their outputs in ways appropriate to the output and to the scale and level of external interest. Effective communication of messages to external stakeholders to facilitate their decision making will form an important element of this support process.

Task 2.D2: Review current barriers to full implementation in industry (BRITEST, with input from industry partners and stakeholders, M5-8)
BRITEST will lead work to look at specific case studies to document the barriers to widespread industrial uptake. The aim is to improve understanding of hurdles that successful projects have had to cross, and reasons why some projects have had more limited industrial uptake.

Participation per Partner

Partner number and short name	WP2 effort
1 - BRITEST	
2 - INEOS	
3 - ASPIRE	
4 - IRIS	
5 - LEEDS	
6 - CRIT	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D2.1	Review of metrics and other good practice for measuring project progress	5 - LEEDS	Report	Confidential, only for members of the consortium (including the Commission Services)	3
D2.2	Initial review of highlevel SPIRE and case study project outputs	1 - BRITEST	Report	Confidential, only for members of the consortium (including the Commission Services)	3

D2.3	Review of approaches to structuring information for educational purposes	5 - LEEDS	Report	Confidential, only for members of the consortium (including the Commission Services)	3
D2.4	Review of available technology scanning approaches	6 - CRIT	Report	Confidential, only for members of the consortium (including the Commission Services)	6
D2.5	Review of best practice decision making on project outputs	5 - LEEDS	Report	Confidential, only for members of the consortium (including	7

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
				the Commission Services)	
D2.6	Review of barriers to industrial implementation	1 - BRITEST	Report	Confidential, only for members of the consortium (including the Commission Services)	8
D2.7	Review of high-level outputs for SPIRE projects	1 - BRITEST	Report	Confidential, only for members of the consortium (including the Commission Services)	13
D2.8	Draft methodology for impact assessment and enhancement	5 - LEEDS	Report	Confidential, only for members of the consortium (including the Commission Services)	13

Description of deliverables

D2.1 – Review of metrics and other good practice for measuring project progress (BRITEST/LEEDS – M3)
D2.2 – Initial review of high-level SPIRE and case study project outputs (BRITEST – M3)
D2.3 – Review of approaches to structuring information for educational purposes (LEEDS/BRITEST – M3)
D2.4 – Review of available technology scanning approaches (CRIT/LEEDS – M6)
D2.5 – Review of best practice decision making on project outputs (LEEDS – M7)
D2.6 – Review of barriers to industrial implementation (BRITEST/ LEEDS – M8)
D2.7 – Review of high-level outputs for SPIRE projects (BRITEST – M13)
D2.8 – Draft methodology for impact assessment and enhancement (LEEDS/BRITEST – M13)

D2.1 : Review of metrics and other good practice for measuring project progress [3]

Report summarising research into good practice for methods and metrics used to measure project progress and impact in SPIRE and beyond. Research to also include how impact across projects is monitored by the Public-PrivatePartnership (SPIRE) and the Commission.

D2.2 : Initial review of high-level SPIRE and case study project outputs [3]

Summary report of the high-level outputs from case study projects.

D2.3 : Review of approaches to structuring information for educational purposes [3]

Summary report of research into approaches used for structuring information (ie project outputs) for educational purposes. To include summary of learning from Educate-to-Innovate.

D2.4 : Review of available technology scanning approaches [6]

Summary report of research into available technology scanning approaches relevant to SPIRE projects.

D2.5 : Review of best practice decision making on project outputs [7]

Summary reporting of research into best practice approaches for making decisions regarding project outputs. Particular focus is given to approaches used in industry.

D2.6 : Review of barriers to industrial implementation [8]

Summary report of issues and barriers faced by industry in adopting outputs from collaborative projects. Research to include survey of SPIRE industry members and other industry stakeholders.

D2.7 : Review of high-level outputs for SPIRE projects [13]

Building of D2.2, this report summarises the high-level outputs from a broader range of SPIRE projects, beyond the primary case studies of SPRING.

D2.8 : Draft methodology for impact assessment and enhancement [13]

Bringing together the research done within WP2, this deliverable summarises a draft proposal for a methodology for impact assessment and enhancement of SPIRE projects (to be tested on case study projects).

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**
MS1	Metrics selected to test on case study projects	5 - LEEDS	4	List of metrics to be agreed with Work Package Leaders.
MS2	Technology scanning methods selected to test on case studies	6 - CRIT	5	List of methods to be agreed with Work Package Leaders.

Work package number [9]	WP3	**Lead beneficiary** [10]	4 - IRIS
Work package title	Testing and Validation		
Start month	1	**End month**	21

Objectives

1. To establish an efficient and high quality methodology, carefully selecting and recruiting the SPIRE projects and casestudies that will be involved during the project execution.
2. To define guidelines for the iterative testing process, in order to ensure a standardised approach to facilitate thedrawing of robust comparisons and conclusions.
3. To implement and manage the testing process, its analysis and documentation

Description of work and role of partners

WP3 - Testing and Validation [Months: 1-21]
IRIS, BRITEST, INEOS, CRIT
Task 3.A3: Phase one testing of good practice project metrics on case studies (IRIS, with input from INEOS & CRIT, M4-8)
A preliminary activity will be to draw up the methodology for carefully selecting and recruiting the SPIRE projects and case studies that will be involved during the project execution will be performed. Projects will be recruited in order to ensure that collectively they target the majority of the sectors of SPIRE.
IRIS will lead the work to implement the first phase of the testing for the metrics obtained within WP2 on the selected case studies. To this end, relevant 'Champions' (i.e. managerial contacts) will be appointed in each selected SPIRE project to ensure that we have an influential link to each project. Our Champions will: ensure active engagement with our project; have sufficient knowledge and insights into their own project; promote changes and novel methodologies in their projects; have the ability to evaluate that the parameters defined are crucial to reach success on the implementation of their own projects. This first phase the testing task will be performed via interviews and using carefully designed surveys to be completed by relevant stakeholders.
The results obtained in this testing phase will feed back into WP2 for refining the project metrics selected but also to WP4 in order to start developing the A4 Task and develop the Preliminary Metrics Guidance. Output: D3.1 (M10)

Task 3.A6: Phase two testing of good practice project metrics on case studies (IRIS, with input from INEOS & CRIT, M13-20)
During the second year and based on the feedback received from WP2 during the first testing phase, IRIS will lead the work to implement a second testing phase in order to improve the definition of the initial results and to extend the data obtained from the industrial partners.
The output produced from this task will be much more accurate and will feed into WP4 A7 task for the generation of the final recommendations and gaps report. Output: D3.2 (M21)

Task 3.B3: Provide high-level project outputs and test the categorisation frameworks (IRIS, with input from INEOS & CRIT, M1-12)
IRIS will lead the work to make an analysis of each project case study objectives and translate them into understandable information for the overall industrial community representing each sector. Taking into consideration the overall outputs from the task B1 in WP2, all this data will be included in a categorisation framework in order to be transferred to WP4 and WP5, where it will be used to develop the framework for education and training resources to be shared publicly. In order to implement this task, a workshop with relevant researchers (academics, industrialists, researchers) from the selected projects will be undertaken at the initial stage of the project to extract the industrial potential benefits from each case study.

Task 3.B6: Review deeper level project outputs from select case study projects (BRITEST, with input from IRIS & INEOS, M5-11)
BRITEST will lead a task to establish which outputs from select case studies could be used as source materials for educational and training resources. Output: D3.4

Task 3.C2: Test technology scanning best practice on case studies (CRIT, with input from IRIS & INEOS, M6-8) CRIT will lead a task to test best practice technology scanning methods (from Task 2.C1) on selected case study projects.

Participation per Partner

Partner number and short name	WP3 effort
1 - BRITEST	
2 - INEOS	
4 - IRIS	
6 - CRIT	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D3.1	Case studies testing report 1	4 - IRIS	Report	Confidential, only for members of the consortium (including the Commission Services)	10
D3.2	Case Studies Testing - Final Report	4 - IRIS	Report	Confidential, only for members of the consortium (including the Commission Services)	21
D3.3	Summary report of case study project outputs suitable for educational/ training resources	1 - BRITEST	Report	Confidential, only for members of the consortium (including the Commission Services)	12

Description of deliverables

D3.1 – 1st report of the results obtained during the testing activities performed in the different case Studies (IRIS – M10)
D3.2 – Final report of the results obtained during the testing activities performed in the different case Studies (IRIS – M21)
D3.3 – Summary report of case study project outputs suitable for educational/ training resources (BRITEST – M12)

D3.1 : Case studies testing report 1 [10]

1st report of the results obtained during the testing activities performed in the Case Studies.

D3.2 : Case Studies Testing - Final Report [21]

Final report of the results obtained during the testing activities performed in the Case Studies

D3.3 : Summary report of case study project outputs suitable for educational/ training resources [12]

Summary report of outputs from case study projects that could be used as source material for the generation of educational/ training resources.

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS1	Metrics selected to test on case study projects	5 - LEEDS	4	List of metrics to be agreed with Work Package Leaders.
MS2	Technology scanning methods selected to test on case studies	6 - CRIT	5	List of methods to be agreed with Work Package Leaders.
MS3	Translated outputs matrix developed for specific and/or cross-sector usage	1 - BRITEST	6	Milestone will be reached through Deliverable 4.1
MS4	Metrics case study testing complete	4 - IRIS	20	Milestone will be reached through completion of D3.2

Work package number [9]	WP4	**Lead beneficiary** [10]	1 - BRITEST
Work package title	Output Generation		
Start month	4	**End month**	24

Objectives

The objective of WP4 is to take research from WP2 and testing experience from WP3 to develop the key exploitable outputs for the project:

A Guidance on how to measure progress/ impact/ success of projects
B Framework for getting different levels of project results to the right audience
- High, categorized level for getting targeted awareness through guided framework on the spire2030.eu portal
- Deeper level for extracting output for training/ teaching resources
C Model for mapping project outputs against industry needs
- Through thematic interactive industry workshops, expert input and technology scanning methods
D Guidance for understanding business barriers to uptake
- Includes best practice for enabling good decision making when evaluating project outputs (include risk and visualisation)
E - Training workshop and presentation package for communicating value and impact with Commission, Industry andbroader stakeholders
F Policy gaps and future SPIRE needs recommendations report

Description of work and role of partners

WP4 - Output Generation [Months: 4-24]
BRITEST, INEOS, ASPIRE, IRIS, LEEDS, CRIT
Task 4.A4: Develop preliminary guidance for best practice measurement of progress/ impact/ success of projects (BRITEST lead, All Partners, M9-11)
BRITEST will work with project partners, bringing the learning from activities A1-A3 to make preliminary guidance recommendations on best practice for measuring process, impact and success of SPIRE projects. Output: D4.2

Task 4.A5: Develop final guidance for best practice measurement of progress/ impact/ success of projects (BRITEST lead, All Partners, M21-23)
BRITEST will work with project partners to update the preliminary guidance (D4.2) to make a final guidance recommendation for the measurement of process, impact and success of SPIRE projects. Output: D4.8

Task 4.B2: Develop translated outputs matrix for specific and/or cross-sector usage (BRITEST lead, input from CRIT, M4-6)
BRITEST will work with CRIT to develop a framework for capturing high-level learning from SPIRE projects so that people from across the SPIRE sectors can more easily find projects of potential interest. Selected case study projects will be used to start populating the framework. Output: D4.1

Task 4.B7: Develop framework for capturing education/ training resources (BRITEST lead, input from LEEDS and CRIT, M9-11)
BRITEST will lead an activity to develop a framework for capturing resources from SPIRE projects so they can be of use to academia and training providers. Information from selected case study projects will be used to provide examples. Output: D4.3

Task 4.B8: Develop final guidance for framing outputs from SPIRE projects (BRITEST lead, All Partners, M18-20)
BRITEST will lead the activity to provide guidance on framing high-level and deeper findings from SPIRE projects to make them more accessible to those in SPIRE industries. This will provide the specification for improvements to the spire2030.eu portal. Output D4.7

Task 4.C3: Develop preliminary thematic mapping guidance for mapping industry needs to project outputs (BRITEST lead, input from CRIT, M9-15)
BRITEST will work with CRIT to develop the structure for a test thematic workshops aimed at (for example) the ceramics sector. This task will take learning from the case study projects translation activities and the trial of technology scanning methods (Task 3.C2) to test how learning from a number of different projects can be brought together to target specific issues for industry, including SME's. The workshops will be delivered through WP5 (Task 5.C5). Output: D4.4

Task 4.C4: Refine thematic mapping guidance (BRITEST lead, All Partners, M16-20)
BRITEST will lead the activity taking the learning from the thematic workshops to develop guidance for running crossproject interactive workshop, which map findings from SPIRE projects and technology scanning to industry needs. Output D4.7
Task 4.E3: Develop training and presentation package for communicating value and impact (BRITEST lead, input from CRIT & ASPIRE, M16-24)
BRITEST will lead the activity to develop a training and presentation package to be available as a SPIRE resource to help those new to SPIRE projects, project coordinators, dissemination/exploitation leads and sustainability practitioners. The package will highlight elements of the SPRING project guidance and best practice. Output: D4.9

Task 4.F1: Develop preliminary recommendations and gaps report (BRITEST lead, All Partners, M17-18)
In line with a M18 review point, BRITEST will lead the activity to pull together preliminary recommendations and gaps from the project. Output: D4.5

Task 4.F2: Develop final recommendations and gaps report (BRITEST lead, All Partners, M22-24)
At the end of the project, BRITEST will lead the task to update the preliminary recommendations and gap report to summarise findings from SPRING. Output: D4.10

Participation per Partner

Partner number and short name	WP4 effort
1 - BRITEST	
2 - INEOS	
3 - ASPIRE	
4 - IRIS	
5 - LEEDS	
6 - CRIT	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D4.1	Structure proposal for capturing high-level findings from SPIRE projects	1 - BRITEST	Report	Confidential, only for members of the consortium (including the Commission Services)	7
D4.2	Preliminary guidance for best practice measurement of progress/ impact/ success of projects	1 - BRITEST	Report	Confidential, only for members of the consortium (including the Commission Services)	13

D4.3	Framework proposal for capturing education/ training resources	1 - BRITEST	Report	Public	12

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D4.4	Preliminary structure for thematic workshops	1 - BRITEST	Report	Confidential, only for members of the consortium (including the Commission Services)	16
D4.5	Preliminary recommendations and gaps report	1 - BRITEST	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D4.6	Guidance for framing SPIRE projects outputs on spire2030.eu	1 - BRITEST	Report	Public	20
D4.7	Thematic workshop structure guidance	1 - BRITEST	Report	Public	21
D4.8	Guidance for best practice measurement of progress, impact and success of SPIRE projects	1 - BRITEST	Report	Confidential, only for members of the consortium (including the Commission Services)	23
D4.9	Training and presentation package for communicating value and impact of SPIRE projects	1 - BRITEST	Websites, patents filling, etc.	Public	24
D4.10	Final recommondations and gaps report	1 - BRITEST	Report	Public	24

Description of deliverables

D4.1 – Structure proposal for capturing high-level findings from SPIRE projects (BRITEST, All – M7)
D4.2 – Preliminary guidance for best practice measurement of progress/ impact/ success of projects (BRITEST, All – M12)
D4.3 – Framework proposal for capturing education/ training resources (BRITEST, LEEDS, CRIT – M12)
D4.4 – Preliminary structure for thematic workshops (BRITEST, CRIT – M16)
D4.5 – Preliminary recommendations and gaps report (BRITEST, All – M18)
D4.6 – Guidance for framing SPIRE projects outputs on spire2030.eu (BRITEST, All – M20)
D4.7 – Thematic workshop structure guidance (BRITEST, All – M21)
D4.8 – Guidance for best practice measurement of progress, impact and success of SPIRE projects (BRITEST, All – M23)
D4.9 – Training and presentation package for communicating value and impact of SPIRE projects (BRITEST, All – M24)
D4.10 – Final recommendations and gaps report (BRITEST, All – M24)

D4.1 : Structure proposal for capturing high-level findings from SPIRE projects [7]

Proposal of structure for capturing high-level findings from SPIRE projects. Report should include potential categories for classifying project outputs.

D4.2 : Preliminary guidance for best practice measurement of progress/ impact/ success of projects [13]
Taking the research summary from D2.1 and D2.8, this deliverable proposes a format for guidance that could be presented to projects, SPIRE and the Commission for best practice measurement and monitoring of progress, impact and success of projects.

D4.3 : Framework proposal for capturing education/ training resources [12]

Proposal for framework for capturing education/ training resources from SPIRE projects.

D4.4 : Preliminary structure for thematic workshops [16]

Structure proposal for thematic workshops to be tested, e.g. on ceramics sector.

D4.5 : Preliminary recommendations and gaps report [18]

Draft summary report of gaps and recommendations found across all SPRING research areas.

D4.6 : Guidance for framing SPIRE projects outputs on spire2030.eu [20]

Final dissemination guidance report for how the spire2030.eu portal should be structured in order to frame outputs from SPIRE projects. This report guides the implementation steps in D5.3.

D4.7 : Thematic workshop structure guidance [21]

Building on D4.4, this report details guidance for structuring and running future thematic workshops.

D4.8 : Guidance for best practice measurement of progress, impact and success of SPIRE projects [23]

Building on D4.2, this deliverable makes a final proposal for improved measuring and monitoring of SPIRE project progress, impact and success. Public dissemination of this guidance is covered in D5.1.

D4.9 : Training and presentation package for communicating value and impact of SPIRE projects [24]

This deliverable takes the learning from the SPRING reports into a training and presentation package that can be used to inform current and future SPIRE projects about best practice approaches for communicating value and impact.

D4.10 : Final recommondations and gaps report [24]

Building on D4.5, this report summarises the remaining gaps and recommendations for enhancing the impact of SPIRE projects.

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**
MS3	Translated outputs matrix developed for specific and/or cross-sector usage	1 - BRITEST	6	Milestone will be reached through Deliverable 4.1
MS5	Frameworks guidance complete	1 - BRITEST	20	Milestone will be reached through completion of D4.6

Work package number [9]	WP5	**Lead beneficiary** [10]	3 - ASPIRE
Work package title	Stakeholder Engagement		
Start month	1	**End month**	24

Objectives

The objectives of WP5 are to ensure broad stakeholder engagement in SPRING activities and to act as the conduit for the project outputs back to SPIRE stakeholders. This encompasses external dissemination and communication activities. WP5 also includes the formation and utilisation of specialist groups to consult with for SPRING guidance and to promote greater collaboration between SPIRE projects to enhance impact once SPRING has completed its work.

Description of work and role of partners

WP5 - Stakeholder Engagement [Months: 1-24]
ASPIRE, BRITEST, INEOS, IRIS, CRIT
Task 5.A5: Incorporate D4.2 into SPIRE guidance (ASPIRE, M12)
Following the development of preliminary guidance for best practice measurement of progress/ impact/ success of projects in WP4, ASPIRE will evaluate and implement relevant guidelines for use by SPIRE projects.

Task 5.A8: Incorporate D4.8 into SPIRE guidance (ASPIRE, M24)
Following the development of final guidance for best practice measurement of progress, impact and success of projects in WP4, ASPIRE will evaluate and update guidelines for use by SPIRE projects. Output D5.1

Task 5.B9: Develop spire2030.eu website portal to incorporate frameworks for capturing SPIRE projects' outputs (ASPIRE, M21-24)
Following the development of frameworks to capture outputs from SPIRE projects in WP4, ASPIRE will lead work to implement framework recommendations on the spire2030.eu website. If considerable development work is required, some modifications may be recommended for implementation beyond the timeframe of SPRING. Output D5.3

Task 5.C5: Run Thematic Workshops (BRITEST, with input from ASPIRE & CRIT, ~M10, M15, M22)
In parallel with the WP4 activities (C4,C5) to develop the structure for thematic workshops, BRITEST, CRIT and ASPIRE will work together to establish and promote a minimum of 2 workshops to help develop and demonstrate the structure. Output: D5.2

Task 5.D3: Establish SME Working Group (ASPIRE, IRIS, Start assembling M1 – fully established by M9-11)
A.SPIRE will work with IRIS to establish a SPIRE SME Working/ Network group to gather input and feedback on the specific challenges facing SMEs in the SPIRE community.

Task 5.E1: Establish SPIRE Coordinators Network Group (ASPIRE, BRITEST, M1-3)
A.SPIRE will work with BRITEST to establish a SPIRE Coordinators Network group to gather input and feedback for SPRING and to enable greater collaboration and information sharing between SPIRE projects.

Task 5.E2: Establish SPIRE Sustainability Practitioners Group (ASPIRE, BRITEST, M1-3)
A.SPIRE will work with BRITEST to establish a SPIRE Sustainability Practitioners group to enable greater collaboration and methodology sharing between SPIRE projects, and to help promote greater impact of some SPIRE project including the 2014:SPIRE4 sustainability tools projects.

Task 5.E4: Incorporate SPRING training package into Spire website (ASPIRE, BRITEST, M24)
A.SPIRE will work with BRITEST to incorporate the training package developed in WP4 into the Spire website portal, to help current and future SPIRE projects.

Task 5.X1: General project communication updates to public and stakeholders (ASPIRE, BRITEST, M1-24)
A.SPIRE and BRITEST will manage external communication activities, including generation and distribution of newsletters, press releases, social media feeds (e.g. Twitter), blogs etc. Timings to coincide with major project deliverables, events and milestones.

Participation per Partner

Partner number and short name	WP5 effort
1 - BRITEST	
2 - INEOS	
3 - ASPIRE	
4 - IRIS	
6 - CRIT	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D5.1	Guidance for project progress, impact and success measurement published	3 - ASPIRE	Report	Public	24
D5.2	Thematic Workshops	1 - BRITEST	Other	Public	22
D5.3	spire2030.eu update to incorporate project output frameworks	3 - ASPIRE	Websites, patents filling, etc.	Public	24
D5.4	Training package on website	3 - ASPIRE	Websites, patents filling, etc.	Public	24

Description of deliverables

D5.1 – Guidance for project progress, impact and success published (ASPIRE, BRITEST – M24)
D5.2 – Thematic workshops (minimum x2) (BRITEST, ASPIRE & CRIT – M22)
D5.3 – spire2030.eu update to incorporate project output frameworks (ASPIRE, BRITEST – M24)
D5.4 – Training package on website (ASPIRE, BRITEST – M24)

D5.1 : Guidance for project progress, impact and success measurement published [24]

This deliverable takes the guidance proposal from D4.8 into a published form, disseminated to the SPIRE community and broader stakeholders.

D5.2 : Thematic Workshops [22]

A minimum of two thematic workshops will be run to test and demonstrate guidance on best practice.

D5.3 : spire2030.eu update to incorporate project output frameworks [24]

Following the development of frameworks to capture outputs from SPIRE projects in WP4, ASPIRE will lead work to implement framework recommendations on the spire2030.eu website. If considerable development work is required, some modifications may be recommended for implementation beyond the timeframe of SPRING.

D5.4 : Training package on website [24]

Deliverable to publish SPRING training package on SPIRE website and provide associated communication to SPIRE stakeholders.

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS5	Frameworks guidance complete	1 - BRITEST	20	Milestone will be reached through completion of D4.6

Work package number [9]	WP6	**Lead beneficiary** [10]	1 - BRITEST
Work package title	Ethics requirements		
Start month	1	**End month**	24

Objectives

The objective is to ensure compliance with the 'ethics requirements' set out in this work package.

Description of work and role of partners

WP6 - Ethics requirements [Months: 1-24]
BRITEST
This work package sets out the 'ethics requirements' that the project must comply with.

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D6.1	NEC - Requirement No. 1	1 - BRITEST	Ethics	Confidential, only for members of the consortium (including the Commission Services)	2
D6.2	POPD - Requirement No. 2	1 - BRITEST	Ethics	Confidential, only for members of the consortium (including the Commission Services)	2

Description of deliverables

The 'ethics requirements' that the project must comply with are included as deliverables in this work package.

D6.1 : NEC - Requirement No. 1 [2]

The applicant must confirm that the ethical standards and guidelines of Horizon2020 will be rigorously applied, regardless of the country in which the research is carried out.

D6.2 : POPD - Requirement No. 2 [2]

Detailed information must be provided on the procedures that will be implemented for data collection, storage, protection, retention and destruction and confirmation that they comply with national and EU legislation.

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification

1.3.4. WT4 List of milestones

Milestone number[18]	Milestone title	WP number[9]	Lead beneficiary	Due Date (in months)[17]	Means of verification
MS1	Metrics selected to test on case study projects	WP2, WP3	5 - LEEDS	4	List of metrics to be agreed with Work Package Leaders.
MS2	Technology scanning methods selected to test on case studies	WP2, WP3	6 - CRIT	5	List of methods to be agreed with Work Package Leaders.
MS3	Translated outputs matrix developed for specific and/or crosssector usage	WP3, WP4	1 - BRITEST	6	Milestone will be reached through Deliverable 4.1
MS4	Metrics case study testing complete	WP3	4 - IRIS	20	Milestone will be reached through completion of D3.2
MS5	Frameworks guidance complete	WP4, WP5	1 - BRITEST	20	Milestone will be reached through completion of D4.6

1.3.5. WT5 Critical Implementation risks and mitigation actions

Risk number	Description of risk	WP Number	Proposed risk-mitigation measures
1	Delay in feeding of data from one WP to other(s)	WP1, WP2, WP3, WP4, WP5	Strict project management according to milestones
2	Inability to successfully engage with/ mobilse ongoing SPIRE projects	WP1, WP2, WP3, WP4, WP5	Effective communication strategy for conveying the value and benefits of engaging with the SPRING project. Support from A.SPIRE in mobilising active SPIRE projects
3	Case study partner withdraws from the consortium	WP3	Resolve solution in the General Assembly, potential compensation by other industrial partners' activities, involvement of new partner (representing a SPIRE case study project) will be considered
4	Poor attendance at stakeholder meetings and thematic workshops	WP5	A.SPIRE has access to sufficiently large pool of stakeholders. Travel budget has been allocated to aid attendance. Timing of workshops and meetings will be set at project inception to give plenty of notice to attendees, and will be set to coincide with other SPIRE meetings, where possible.
5	A partner is unable to carry out their tasks to the project timetable	WP1, WP2, WP3, WP4, WP5	Monthly WP reviews to monitor progress, flexibility to allocate additional assistance from other partners (within resource budget) to meet deadlines. Slack in Gantt to allow minor slippage
6	Low acceptance and uptake of recommendations by industry/ policy makers	WP5	A.SPIRE in position to directly implement many of the recommendations. Industry advisory board to ensure recommendations are useful to industry. Clear recommendations report delivered in WP4 to identify hurdles.

1.3.6. WT6 Summary of project effort in person-months

	WP1	**WP2**	**WP3**	**WP4**	**WP5**	**WP6**	**Total Person/Months per Participant**
1 - BRITEST							
2 - INEOS							
3 - ASPIRE							
4 - IRIS							
5 - LEEDS							
6 - CRIT							
Total Person/Months							

1.3.7. WT7 Tentative schedule of project reviews

Review number [19]	Tentative timing	Planned venue of review	Comments, if any
RV1	18	Brussels	mid-term review
RV2	24	Brussels	final review

1. Project number

The project number has been assigned by the Commission as the unique identifier for your project. It cannot be changed. The project number **should appear on each page of the grant agreement preparation documents (part A and part B)** to prevent errors during its handling.

2. Project acronym

Use the project acronym as given in the submitted proposal. It can generally not be changed. The same acronym **should appear on each page of the grant agreement preparation documents (part A and part B)** to prevent errors during its handling.

3. Project title

Use the title (preferably no longer than 200 characters) as indicated in the submitted proposal. Minor corrections are possible if agreed during the preparation of the grant agreement.

4. Starting date

Unless a specific (fixed) starting date is duly justified and agreed upon during the preparation of the Grant Agreement, the project will start on the first day of the month following the entry into force of the Grant Agreement (NB : entry into force = signature by the Commission). Please note that if a fixed starting date is used, you will be required to provide a written justification.

5. Duration

Insert the duration of the project in full months.

6. Call (part) identifier

The Call (part) identifier is the reference number given in the call or part of the call you were addressing, as indicated in the publication of the call in the Official Journal of the European Union. You have to use the identifier given by the Commission in the letter inviting to prepare the grant agreement.

7. Abstract

8. Project Entry Month

The month at which the participant joined the consortium, month 1 marking the start date of the project, and all other start dates being relative to this start date.

9. Work Package number

Work package number: WP1, WP2, WP3, ..., WPn

10. Lead beneficiary

This must be one of the beneficiaries in the grant (not a third party) - Number of the beneficiary leading the work in this work package

11. Person-months per work package

The total number of person-months allocated to each work package.

12. Start month

Relative start date for the work in the specific work packages, month 1 marking the start date of the project, and all other start dates being relative to this start date.

13. End month

Relative end date, month 1 marking the start date of the project, and all end dates being relative to this start date.

14. Deliverable number

Deliverable numbers: D1 - Dn

15. Type

Please indicate the type of the deliverable using one of the following codes:

R Document, report
DEM Demonstrator, pilot, prototype DEC
Websites, patent fillings, videos, etc.
OTHER
ETHICS Ethics requirement
ORDP Open Research Data Pilot

16. Dissemination level

Please indicate the dissemination level using one of the following codes:

PU Public
CO Confidential, only for members of the consortium (including the Commission Services)
EU-RES Classified Information: RESTREINT UE (Commission Decision 2005/444/EC)
EU-CON Classified Information: CONFIDENTIEL UE (Commission Decision 2005/444/EC) EU-SEC Classified Information: SECRET UE (Commission Decision 2005/444/EC)

17. Delivery date for Deliverable

Month in which the deliverables will be available, month 1 marking the start date of the project, and all delivery dates being relative to this start date.

18. Milestone number

Milestone number:MS1, MS2, ..., MSn

19. Review number

Review number: RV1, RV2, ..., RVn

20. Installation Number

Number progressively the installations of a same infrastructure. An installation is a part of an infrastructure that could be used independently from the rest.

21. Installation country

Code of the country where the installation is located or IO if the access provider (the beneficiary or linked third party) is an international organization, an ERIC or a similar legal entity.

22. Type of access

VA if virtual access,
TA-uc if trans-national access with access costs declared on the basis of unit cost,
TA-ac if trans-national access with access costs declared as actual costs, and
TA-cb if trans-national access with access costs declared as a combination of actual costs and costs on the basis of unit cost.

23. Access costs

Cost of the access provided under the project. For virtual access fill only the second column. For trans-national access fill one of the two columns or both according to the way access costs are declared. Trans-national access costs on the basis of unit cost will result from the unit cost by the quantity of access to be provided.

PROJECT SPRING

– SETTING THE FRAMEWORK FOR ENHANCED IMPACT OF SPIRE PROJECTS

Response to:

Call*: H2020-IND-CE-2016-17*

Topic*: SPIRE-11-2017* – **Support for the enhancement of the impact of SPIRE PPP projects**

ANNEX 1 TO THE GRANT AGREEMENT (DESCRIPTION OF THE ACTION)

History of Changes

Version	Date	Changes
2.0	27-Apr-2017	- Confirmation of data management requirements for third country entities (p23) - Correction to timespan of WP2 and WP3, ending in M13 and M21 respectively to coincide with completion of deliverables reporting (Part A) – note, actual research and trial work is still planned to complete M12 and M20.
3.0	20-Jun-2017	- Re-classification of subcontracting costs as Other Direct Costs. Movement of INEOS Travel and Subsistence costs from coordinator budget to INEOS.

CONTENTS

EXECUTIVE SUMMARY 4

1. EXCELLENCE 6

1.1 OBJECTIVES 6
1.2 RELATION TO THE WORK PROGRAMME 7
1.3 CONCEPT AND METHODOLOGY, QUALITY OF THE COORDINATION AND SUPPORT MEASURES 10
1.3.1 – Principals and Concept 10
1.3.2 – Approaches and Activities ..... 11 Project SPRING mapping of Key Activities to Outputs and Objectives 12
1.3.3 – Links to other Innovation Activities 15
1.3.4 – Coordination and Management 18
1.3.5 – Gender 18
2. IMPACT 19
2.1 EXPECTED IMPACTS 19
2.1.1 – Contribution to the expected impacts 19
2.1.2 – Barriers and Framework Conditions 20
2.2 MEASURES TO MAXIMISE IMPACT 21
2.2.1. Dissemination and exploitation of results 21
2.2.2. Communication activities 24
3. IMPLEMENTATION 27
3.1 WORK PLAN – WORK PACKAGES AND DELIVERABLES 27
Project SPRING – High Level Timeline – Year 1 28
Project SPRING – High Level Timeline – Year 2 29
3.2 MANAGEMENT STRUCTURE AND PROCEDURES 30
3.3 CONSORTIUM AS A WHOLE 32
3.4 RESOURCES TO BE COMMITTED 33
Table 3.4 b – 'Other direct cost' items (travel, equipment, infrastructure, goods and services) 33

EXECUTIVE SUMMARY

Background

Increasing industrial uptake of project findings is at the heart of project SPRING. It is the essential building block for ensuring greater impact of SPIRE projects and therefore progress towards the SPIRE roadmap goals of increased resource and energy efficiency in the EU process industries.

Instead of focusing on a small cluster of projects, SPRING has been developed to provide the mechanism to enhance the impact of *all* SPIRE projects.

Objectives

Project SPRING's objective is to increase progress towards the SPIRE goals and enhance return on investment in projects by *addressing the needs of those who make the decisions to adopt process innovations in industry and barriers to their adoption*. It will do this by providing guidance to project participants, decision makers in industry and broader SPIRE stakeholders, enabling them collectively to:

1. Improve the articulation of the value of project exploitable outputs
2. Improve the articulation of industry needs and barriers-to-uptake of exploitable outputs
3. Improve the mapping of project value to industry needs
4. Identify policy gaps and recommendations to improve project impact

Approach

To address these objectives, the project will deliver six sets of exploitable outputs:

A – Guidance on best practice of how to measure progress, impact and success of SPIRE projects.

B – Frameworks for getting different levels of project results to the right audience through the spire2030.eu portal

C – A model for mapping project outputs to industry needs, through thematic, interactive industry workshops, expert input and technology scanning methods.

D – Guidance for understanding business barriers to uptake, including best practice for enabling good decision making when evaluating project outputs.

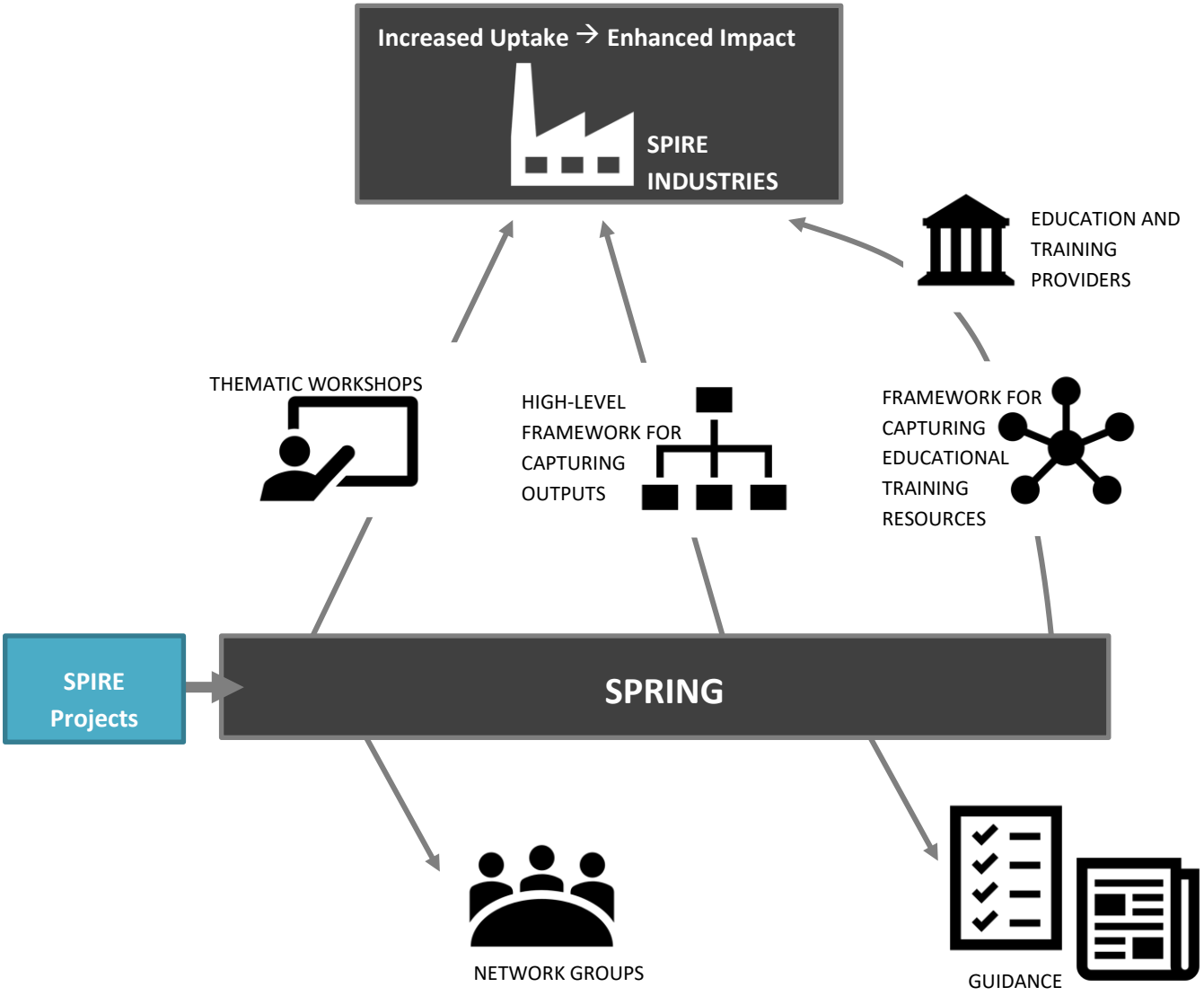
E – A package of training and network groups to upskill SPIRE project participants

F – Identification of policy gaps and future SPIRE needs

The SPRING consortium of six partners represent seven SPIRE projects, which will be used as case studies: **ProPAT**, **Sharebox**, **IbD**, **EPOS, COPRO, STYLE,** and **DREAM.** SPRING will develop the exploitable outputs through a series of work streams that research best practice; consult with stakeholders; test and validate the case studies; generate guidance and frameworks; and disseminate back to the SPIRE community.

The SPRING consortium is industry biased, with an additional Industry Advisory Board to further ensure that the work is focused on the barriers to industrial uptake. Through the inclusion of A.SPIRE in the consortium, SPRING will help direct the existing resources of the PPP, to work more effectively for its members, the Commission and stakeholders, though its communications, website portal and events.

Working across the whole innovation eco-system, SPRING will ensure that outputs from **all SPIRE projects** can achieve their full potential for industrial uptake within the European process industries.

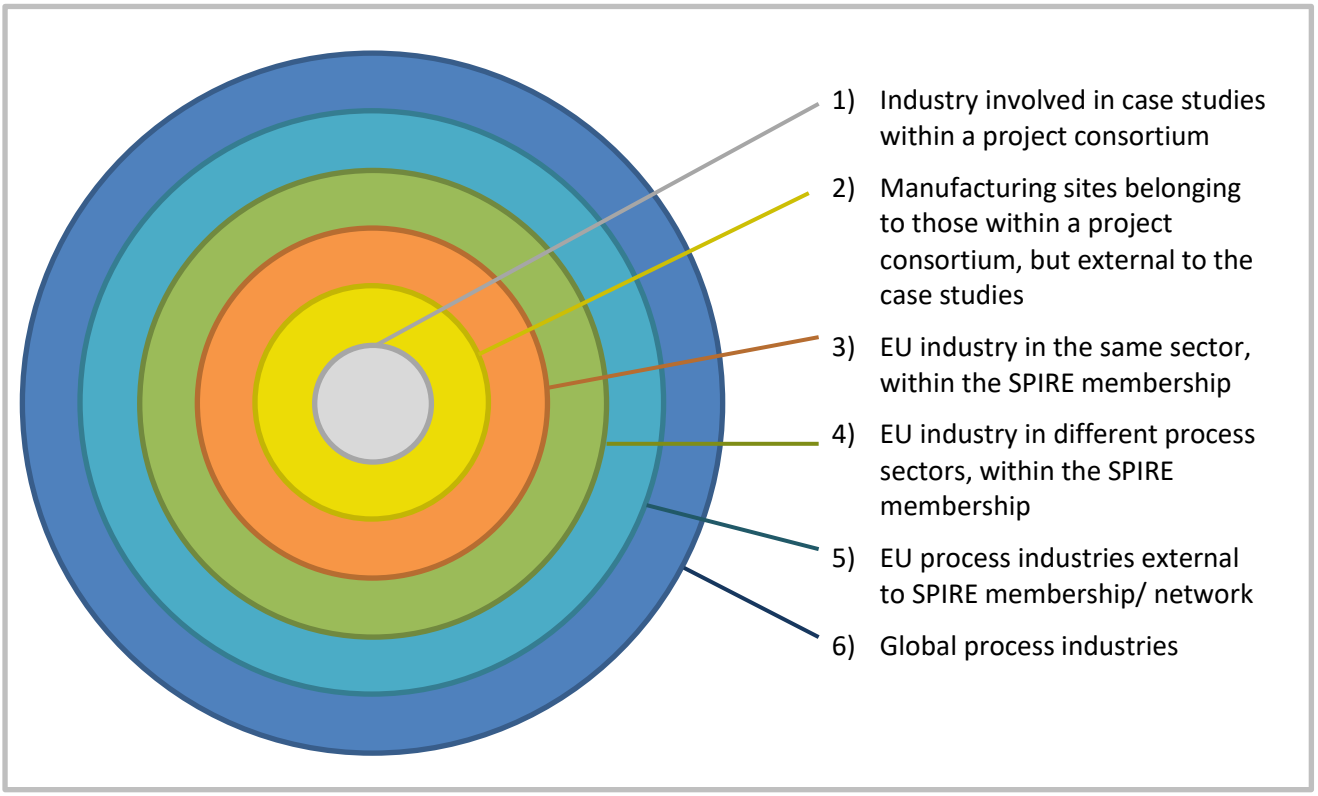


1. EXCELLENCE

1.1 OBJECTIVES

There is potential to increase the impact of collaborative projects within and beyond the SPIRE Public Private Partnership.

Collaborative projects aim to get their findings adopted by industry at various levels:



Each of these levels involves a different set of barriers to uptake (which are developed in Section 1.3.1).

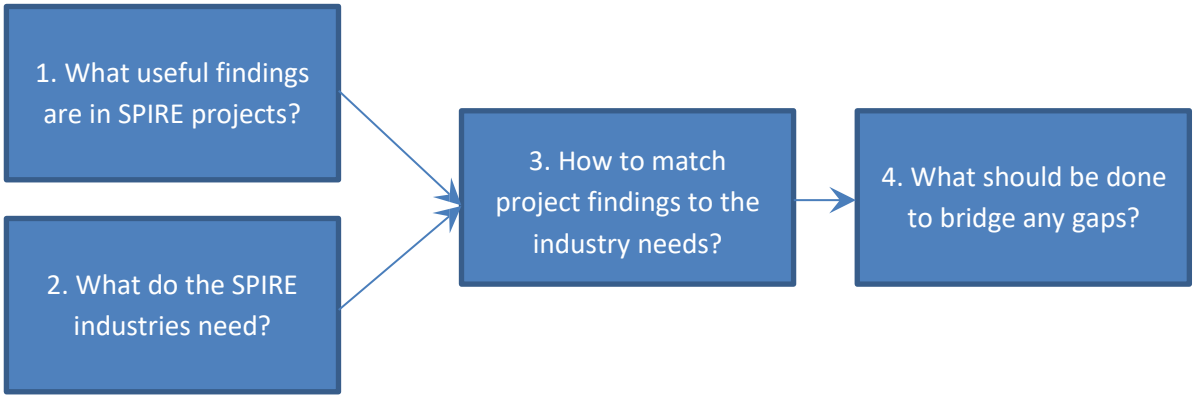
Some recommendations from projects may be implemented by organisations other than industrial manufacturers, i.e. academia, Research and Technology Organisations (RTOs), policy makers. Successful exploitation, however, should be primarily measured by industrial uptake, as this is how SPIRE can meet its vision for resource and energy efficiency improvements[1]. Other organisations are enablers of industrial exploitation and impact.

Consequently, **Project SPRING's objective is to increase progression towards the SPIRE goals and enhance project return on investment by *addressing the needs and barriers of those who make the decisions to adopt process innovations in industry*. It will do this by providing guidance to project participants, decision makers in industry and broader SPIRE stakeholders to:**

1. **Improve the articulation of the value of project exploitable outputs**
2. **Improve the articulation of industry needs and barriers-to-uptake of exploitable outputs**
3. **Improve the mapping of project value to industry needs**
4. **Identify policy gaps and recommendations to improve project impact**

This can be simplified as:

[1] https://www.spire2030.eu/what/walking-the-spire-roadmap/spire-vision



Project SPRING aims to tackle these questions to provide a better structure for *enhancing the impact of all SPIRE projects*, with the aid of a sub-set of Case Study SPIRE projects, representatives of which make up the SPRING project consortium.

1.2 RELATION TO THE WORK PROGRAMME

Project SPRING addresses the topic "Support for the enhancement of the impact of SPIRE PPP projects".

Sustainable Process Industry through Resource and energy Efficiency (SPIRE) is a Public Private Partnership which aims to contribute to the rejuvenation, growth and sustainability of the European Process Industries. Members of SPIRE represent process industries such as cement, ceramics, chemicals, engineering, minerals and ores, non-ferrous metals, steel and water. Together these sectors account for 20% of total European industry as measured by both turnover and employees.

The first projects from the SPIRE PPP came out of 2014 calls, with 53 projects funded to-date. The following table details a selection of the 2014-2016 projects and their '*Technical Field*':

Technical Field	Project Title	Full Title	Call year	Duration -months
Sustainability Tools	STYLE	Sustainability Toolkit for easY Life-cycle Evaluation	2014	24
Sustainability Tools	SAMT	Sustainability assessment methods and tools to support decision-making in the process industries	2014	24
Sustainability Tools	MEASURE	Metrics for Sustainability Assessment in European Process Industries	2014	15
Process Control	CONSENS	Integrated Control and Sensing for Sustainable Operation of Flexible Intensified Processes	2014	36
Process Control	icspec	in-line Cascade laser spectrometer for process control	2014	36
Process Control	ProPAT	Robust and affordable process control technologies for improving standards and optimising industrial operations	2014	48

Process Control	RECOBA	Cross-sectorial real-time sensing, advanced control and optimisation of batch processes saving energy and raw materials	2014	36
Process Control	DISIRE	Integrated Process Control based on Distributed In-Situ Sensors into Raw Material and Energy Feedstock	2014	36
Downstream Processing	PRODIAS	PROcessing Diluted Aqueous Systems	2014	48
Solids handling	IbD	Intensified by Design® for the intensification of processes involving solids handling	2015	36
Energy and resource management systems	EPOS	Enhanced energy and resource Efficiency and Performance in process industry Operations via onsite and cross-sectorial Symbiosis	2015	48
Energy and resource management systems	MAESTRI	Total resource and energy efficiency management system for process industries	2015	48
Energy and resource management systems	SHAREBOX	Secure Management Platform for Shared Process Resources	2015	48
Energy and resource management systems	SYMBIOPTIMA	Human-mimetic approach to the integrated monitoring, management and optimization of a symbiotic cluster of smart production units	2015	36
…				
Circular economy	BAMB, CABRISS, FISSAC, RESLAG, RESYNTEX		2014	
Use of renewables	MefCO2, MOBILE FLIP, SteamBIO		2014	
Process Intensification	ADREM, MEMERE, PRINTCR3DIT, ROMEO, TERRA		2015	
Recovery Technologies	ADIR, REE4EU, REMAGHIC		2015	
Use of heat recovery	I-ThERM, Indus3Es, SUSPIRE, TASIO		2015	
Industrial Furnace Design	DREAM, IMPROOF, VULKANO		2016	
Water management systems	INSPIREWATER, ReWaCEM, SPOTVIEW		2016	
Valorisation of waste heat	DRYficiency, Smartrec		2016	
Valorisation of bioresources	Bio4Products, KARMA2020, REHAP		2016	
Use of CO/CO…	CarbonNext		2016	
Plant-wide monitoring	COCOP, CoPro, FUDIPO, MONSOON		2016	
Business Models	INSPIRE		2016	

Table 1 – SPIRE Projects 2014-2016 Grouped by Technical Field

The impact of SPIRE projects can be enhanced by clustering under their high level *technical fields.* As an exemplar, the three SPIRE-04-2014 Sustainability Tools CSA projects – STYLE, SAMT and MEASURE – demonstrated that collaborating across projects can be beneficial for the European Commission (EC), the SPIRE community, stakeholders and the projects themselves. The project coordinators met prior to kick-off to share information on project activities and timelines and to agree focus areas for the different projects. This ensured that stakeholder communications (e.g. press releases, questionnaires, conference talks) were coordinated, had greater reach and impact and avoided confusion; joint events and workshops were held and projects could focus their efforts in different areas, rather than wasting effort, and duplicating work. Given BRITEST's role as

coordinator of STYLE, learning from this clustered approach will be taken into SPRING to advise on additional areas for potential clustering and good practice for successful clustered activities.

Although some clustering of activities can be easily done across projects within the same call, there is further value to be had from clustering activities across projects funded through different calls. By mapping high level outputs from SPIRE projects, SPRING will make it easier to pull together alternative collections of projects, which may not have previously been linked, leading to thematic workshops and linking of outputs from these projects to industry needs. The mapping should also help identify synergies with other H2020 programme themes and topics, such as BBI, Circular Economy, Water, Energy etc.

The SPRING consortium has strong links to existing technology transfer activities and organisations including SME's (e.g. Enterprise Europe Network (EEN), UK Knowledge Transfer Network, regional cluster organisations, European Technology Platforms and National Technology Platforms such as SusChem etc.). Trials will be conducted within SPRING to partner with these organisations for the running of thematic workshops and as a conduit for 'translated' project outputs.

The challenge to *focus on knowledge transfer and training issues regarding present and future industrial workers* is being addressed in SPRING at two different levels:

- Mapping of high level outputs and development of a framework for the spire2030.eu portal provides a mechanism for directing more people towards relevant output from projects
- Drawing from the experiences of the SusChem Educate to Innovate programme[2] and additional input from SPRING academic stakeholders to develop a framework for capturing educational and training resources developed in SPIRE projects on the spire2030.eu website portal

Project SPRING addresses the following Key Actions from the SPIRE Roadmap:

KA 5.1: Identification, benchmarking and cross-sectorial transfer of good energy and resource efficiency solutions and practices

KA 5.3: Develop skills and education programmes required for the development and deployment of novel energy and resource efficiency solutions and practices

KA 5.4: Enhancing innovation and entrepreneurial skills and culture

[2] http://www.suschem.org/priorities/education/educate-to-innovate.aspx, accessed 16/01/2017

1.3 CONCEPT AND METHODOLOGY, QUALITY OF THE COORDINATION AND SUPPORT MEASURES

1.3.1 – PRINCIPALS AND CONCEPT

The principle behind Project SPRING is to tackle the barriers-to-uptake of project findings. At a high level, these barriers can be classified as:

- **Awareness** – industry unaware of project findings; project unaware of other developments happening in parallel
- **Understanding** – project findings difficult to understand or information fragmented, which may be the fault of communication and the ability to effectively tailor communications to the different target audiences
- **Insufficient skills** in industry to evaluate and implement project findings
- **Specificity** – project targeted at niche sector, insufficient **translation** of broader learning and transferability to wider applications
- Findings thought to be irrelevant or **insufficiently innovative**
- **Lack of consistency** in metrics used – difficult to compare or evaluate different technological solutions
- Lack of buy-in from industry **decision makers**
- **Value** perceived does not match effort/expense required (too costly)
- **Risk of 'new'** outweighs low risk implementation of known tech
- Perceived **patent barriers**
- Lack of appropriate **incentives** (policy 'carrots')
- Lack of appropriate **regulations** (policy 'sticks')
- Specific **SME challenges** (both for exploiting innovations on their own plants and getting their technologies into big industry)
- Many public funded projects have been further limited by the dissemination and exploitation tasks only being *fully* developed at the **end of the projects**, by which point, the **funding ceases**.

Consequently, the concept of SPRING is to research, test and deliver six sets of outputs, which tackle the primary barriers-to-uptake:

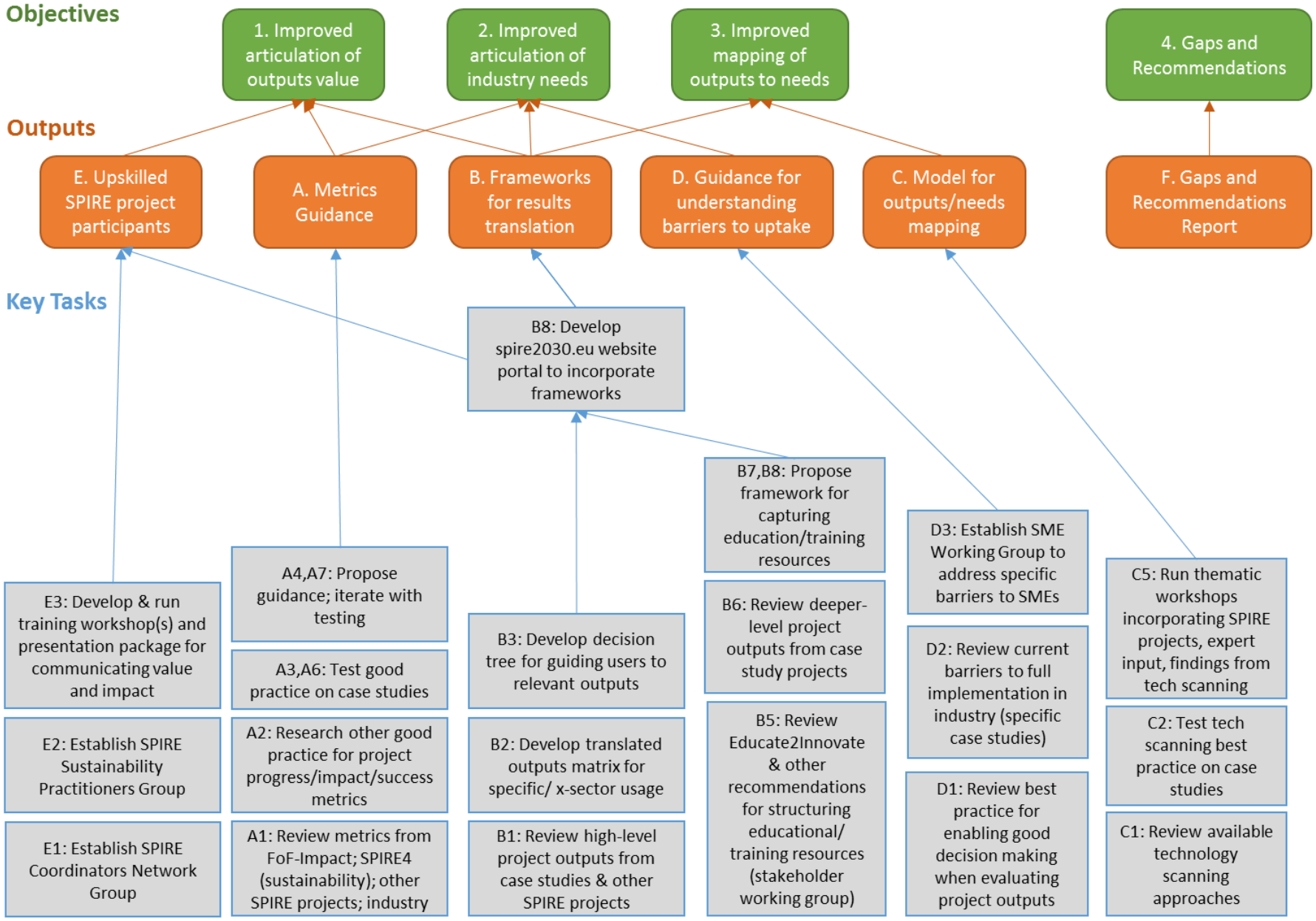
Output Set	Outputs
A	Guidance on how to measure progress/ impact/ success of projects
B	Framework for getting different levels of project results to the right audience - High, categorized level for getting targeted awareness through guided framework on the spire2030.eu portal - Deeper level for extracting output for training/ teaching resources
C	Model for mapping project outputs against industry needs - Through thematic interactive industry workshops, expert input and technology scanning methods
D	Guidance for understanding business barriers to uptake - Includes best practice for enabling good decision making when evaluating project outputs (include risk and visualisation) - SPIRE SME Working Group
E	Upskilled SPIRE project participants, better able to articulate exploitable output value and increase collaboration between projects - Training workshop and presentation package for communicating value and impact with Commission, Industry and broader stakeholders - SPIRE Coordinators Network Group - SPIRE Sustainability Practitioners Working Group
F	Policy gaps and future SPIRE needs recommendations report - Developed through research/ workshop/ other activities

The background to each of these Outputs is detailed in Section 1.3.2.

1.3.2 – APPROACHES AND ACTIVITIES

Project SPRING has been based on a top-down approach; mapping required outputs against overall objectives and the barriers-to-uptake; then plotting activities and tasks to deliver the outputs. An overview of this structure is presented in the following diagram:

PROJECT SPRING MAPPI NG OF KEY ACTIVITIES TO OUTPUTS AND OBJE CTIVES



The Work Packages for SPRING sit in a matrix across the *work streams* to deliver the output sets.

WP1 (Coordination and Management), led by BRITEST, oversees the project and the flow and interrelationships of tasks through the work streams.

WP2 (Research), led by LEEDS, gathers knowledge from partners within the consortium, the SPIRE and FoF projects that they represent, stakeholders in the SPIRE community and broader project and policy research to propose best practice methods to be tested in WP3 (as illustrated in *Figure 1 - Example Work Stream for generating Project SPRING outputs*).

WP3 (Testing & Validation), led by IRIS, takes the recommended methods from WP2, tests their use and value in the case study projects and provides the learning and project perspective into WP4.

WP4 (Output Generation), led by BRITEST, takes the experience from WP3 and research from WP2 to develop the exploitable outputs for SPRING. These include guidance for current and future SPIRE projects, frameworks for the spire2030.eu website and training.

WP5 (Stakeholder Engagement), led by A.SPIRE, provides a route to gather input from the broader SPIRE community and for the outputs and best practice from the project to be widely communicated and disseminated to the stakeholders.

A typical work stream for SPRING, and its interactions across the Work Packages, is shown in the following diagram:

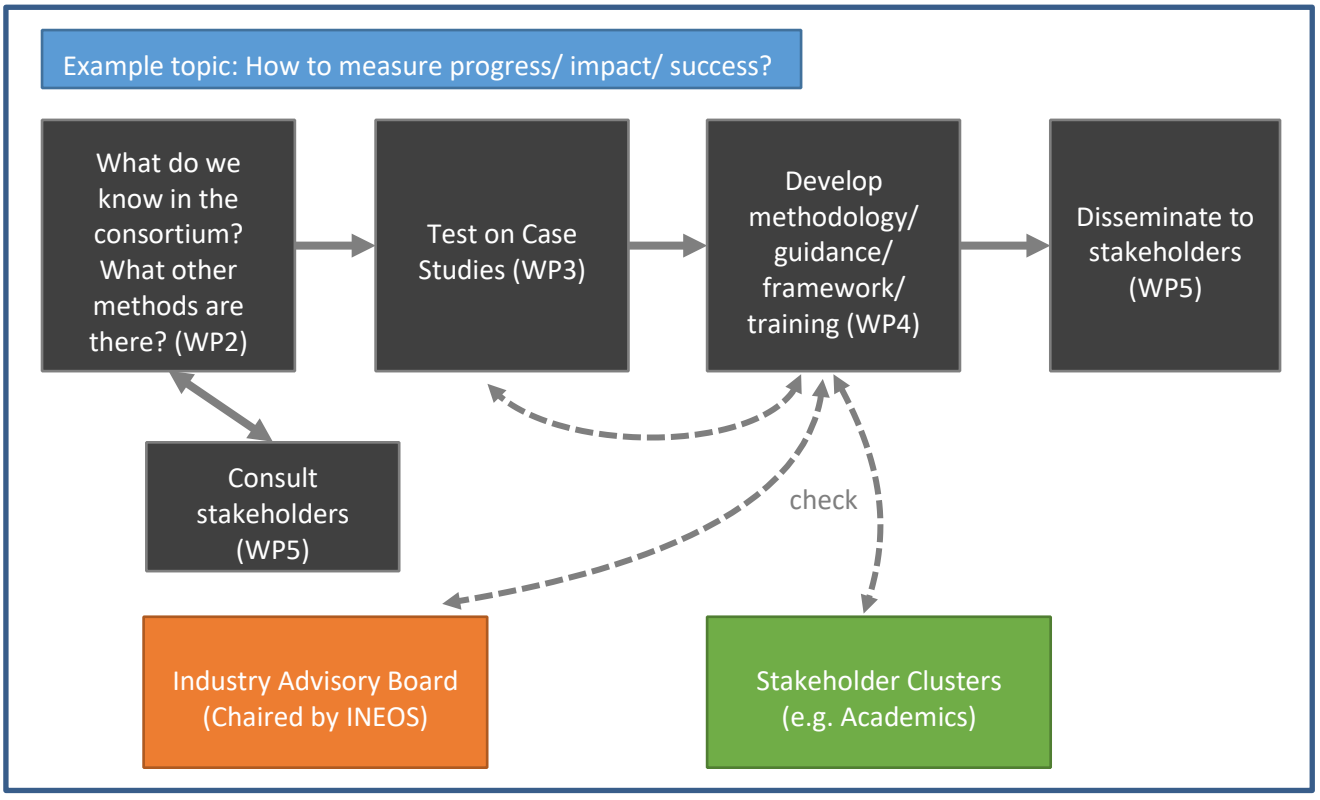


Figure Table 21 – - Example Work Stream for generating Project SPRING outputsExample Work Stream for generating Project SPRING outputs

Output Background

A – Metrics Guidance: Many different metrics are used by projects, SPIRE and the EC to monitor progress on projects, assess their success and their ultimate expected impact. Some metrics are established internally to the projects, others from questionnaires sent out to project partners and industry organisations. For the past two years, Impact Workshops have been organised in Brussels to help collate impact metrics across SPIRE projects. However, these have often demonstrated that the projects measure impact in different ways and attempts to collate mixtures of absolute and percentage metrics can result in meaningless or unrealistic numbers. SPRING aims to review metrics in use across SPIRE projects, those requested from SPIRE and the EC (i.e. the SPIRE KPIs as detailed below), and those used internally by industry, to establish best practice guidelines. This activity will also incorporate recommendations from the SPIRE-04-2014 projects regarding the use of sustainability metrics within SPIRE projects.

SPIRE Key Performance Indicators (KPIs) **at PPP implementation level** are:

- New systems and technologies developed in the relevant sectors
- Participation and benefits for SMEs
- Contribution to the reduction of energy use
- Contribution to the reduction in the use of material resources
- New high-skilled profiles and new curricula developed
- Private investment mobilised in relation to the PPP activities
- Contributions to new standards

SPIRE Key Performance Indicators (KPIs) **at project impact level** are:

- Scale of reduction in energy, material resources and waste
- Project results taken-up for further investments
- Trainings for a higher quality workforce
- Patents and activities leading to standardisation

B – Frameworks for Results Translation: ASPIRE have recently upgraded their website portal to act as a Knowledge Platform for SPIRE members and projects. At present, the platform allows for project search via *Technical Field* and soon to be introduced *key-word search*. SPRING partners will investigate and develop best practice approaches for structures to enable outputs from SPIRE projects to be *more accessible* for two specific audiences:

i) Industry looking for opportunities to improve their manufacturing operations **ii)** Education and training providers looking for source material from innovative projects

For audience (i), the framework is intended to use a decision tree structure to direct users to projects which they may not have previously been aware of. Instead of users having to be proactive in reviewing a large list of project abstracts, or lucky enough to have selected a suitable keyword, the framework should allow genericised outputs from projects to be linked to the characteristics of the industry manufacturing process (e.g. innovations relating to specific operating temperature ranges, or materials of construction, or processing scales, or addressing specific control issues etc.).

For audience (ii), the framework structure will contain a minimum standardisation threshold in term of quality, format and content.

SPRING will work with Case Study projects to see how much translation of their existing outputs is required to meet the frameworks' needs. Guidance will also be produced to allow current and future projects to structure their outputs to fit into the proposed frameworks.

C – Model for Outputs/ Needs Mapping: Intentions have been expressed for SPIRE to run thematic workshop to bring different SPIRE projects together with stakeholders[3]. Instead of just starting with a theme and getting a group of projects to present their outputs in a seminar format, SPRING partners have planned a series of activities that will *culminate* in thematic workshops. For a workshop to be successful in increasing industrial uptake of project outputs, it needs to map the needs of industry with the selected outputs from projects. Impact can also be enhanced by framing the outputs in the context of technological advances that are occurring in parallel to the SPIRE projects; hence SPRING will also review techniques used within projects for 'technology scanning', aiming to incorporate outputs also within the thematic workshop format.

D – Guidance for Understanding Barriers to Uptake: Linked to Output C, SPRING will explore both generic and specific Case Study barriers that have limited industrial uptake of project outputs. This work will lead to the development of guidance to assist current and future SPIRE projects. Specific focus will be put on the challenges facing SME's through the formation of an SME Working Group.

E – Upskilled Project Participants: Project SPRING will form network groups, helping gather input from projects and providing a route to SPRING project dissemination. The use of such network groups should also ensure that SPRING training materials are seen as collaborative sharing of best practice, rather than one project instructing other experienced project participants how to perform their tasks. The training packages will be developed to meet gaps in open access training relevant to SPIRE projects. Topics that may be addressed include Business Opportunity Planning (using Lean Model Canvas[4] and other tools); Communicating Impact; Integrating Sustainability into Project Management etc. Where relevant,

SPRING will also incorporate or reference existing EC advice to project participants (e.g. Executive Agency for SMEs – Project Communication Advice https://ec.europa.eu/easme/en/communicationtoolkit).

1.3.3 – LINKS TO OTHER INNOVATION ACTIVITIES

This SPIRE-11 call requires a collaborative approach, cutting across sectors and industries. The SPRING project will therefore call upon, and feed into, several past, current and future initiatives across Europe funded either by European FP 6&7 programmes, H2020, directly by industry or industry bodies. It is anticipated that a definitive list of related activities will be drawn up as part of the project inception, but examples of known activities that will be particularly relevant are:

SusChem Educate to Innovate – Britest was a partner in this SusChem programme which sought to exploit innovation outputs from SusChem's Research and Innovation projects in order to enhance the innovation skills of future generations of scientists and engineers through the effective engagement of industry and higher education institutions[5]. Learning will be taken from this programme to help direct the development of a structure to capture SPIRE projects' educational and training resources. An example of the learning is the feedback from academic workshops that were held in Educate to

[3] SPIRE AGM, Brussels, November 2016

[4] https://leanstack.com/businessmodelcanvas/

[5] http://www.suschem.org/documents/document/20120420111017-educate_to_innovate_final_web.pdf

Innovate, which showed that it wasn't feasible to develop material that would fit directly into all chemistry/ chemical engineering courses, given the different course structures used in academic institutions across the EU; however, material could be formatted in certain ways to make it more useful for those in academic institutions to incorporate into their courses as well as Life Long Learning programmes.

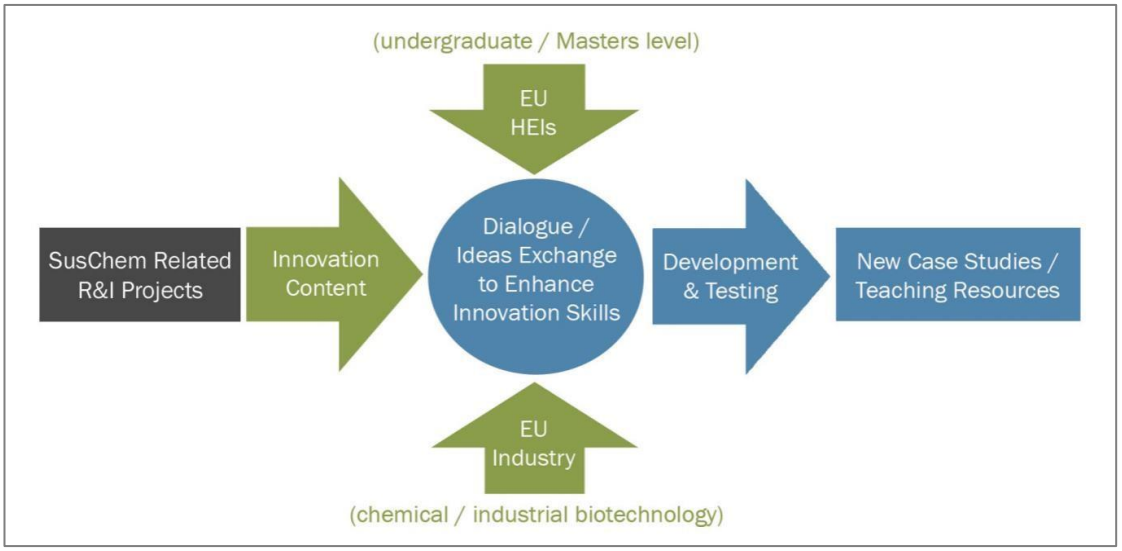


Figure 2 – Educate to Innovate Methodology

Factories of the Future Enhancing Impact CSA's – The FoF-07-2014 call, *Support for the enhancement of the impact of FoF PPP Projects* led to the funding of five projects. One of these projects was '*FoFImpact',* in which CRIT was a partner. Consequently, SPRING will take the learning from this project, along with the other FoF CSA's to ensure that SPRING is starting from a higher foundation for awareness of best practice for impact enhancement within H2020 projects. Examples of learning from FoF-Impact are:

- The need to address advice and exploitation-enhancing activities to all phases of a project lifecycle, from proposal writing through project implementation, and to after the funded action has ended.
- The tendency of project partners to pursue exploitation activities individually rather than at consortium level, due to the different business models present within a consortium, and to the possible issues related to sharing revenues from project results.
- The fact that projects funded under PPPs such as SPIRE and FoF are normally already industrydriven, thus having quite clear exploitation pathways. This results in possible conflicts when expanding exploitation investment beyond what has been already planned (e.g. starting a new company to commercialise results that a large industrial group would prefer to keep for themselves)

SPIRE-12-2017 Call Projects – Projects that will address standardization needs and ways to overcome regulatory bottlenecks in the process industry.

SPIRE projects represented by the SPRING consortium, which will be used as Case Studies for the testing of best practice, frameworks and guidance:

ProPAT – Robust and affordable process control technologies for improving standards and optimising industrial operations. The ProPAT project funded through the SPIRE-01-2014 - Integrated Process Control call, which is being *coordinated by IRIS*, aims to develop robust and affordable process control technologies for improving standards and optimising industrial operations. Over the course of the four years of this project, state-of-the art sensors and analysers that can improve the control and efficiency of the process industry will be tested, optimised and integrated in a centralised Global System Control Platform and validated in processing applications to demonstrate the feasibility of ProPAT as an integrated process control solution for future wide European deployment in the process industry. The R&D effort will centre on engineering design, software/algorithm design, data analytics and data management, process engineering and industrial integration and validation.

Sharebox – Secure Management Platform for Shared Process Resources. The Sharebox project was funded in SPIRE-06-2015 - Energy and resource management systems for improved efficiency in the process industries*. Coordinated by IRIS*, this project aims to pave the way forward for industrial symbiosis as a "solution for more efficient processing and energy systems for the process industry". The SHAREBOX project is developing a secure software platform for the flexible management of shared process resources that will provide plant operations and production managers with the robust and reliable information that they need in real-time in order to effectively and confidently share resources (plant, energy, water, residues, and recycled materials) with other companies in an optimum symbiotic eco-system.

IbD - Intensified-by-Design (IbD®). The IbD project*, led by IRIS*, was funded in SPIRE-08-2015 - Solids handling for intensified process technology. Following an innovative intensified-by-design® (IbD) approach, this Innovation Action project is developing a holistic software platform for facilitating process intensification design and optimisation in processes in which solids are an intrinsic part. Through six IbD®- enabled industrial process intensification case studies, the project will also develop and upgrade process intensification (PI) methods for the handling of solids in continuous production units based, on the one hand, on the intensification of currently existing PI processes and, on the other hand, through completely new PI approaches to the processing of solids.

EPOS (H2020 SPIRE-06-2015): Enhanced energy & resource Efficiency and Performance in process industry Operations via onsite and cross-sectorial Symbiosis. This project, *coordinated by UGent with INEOS as partner*, brings together 5 global process industries from 5 key relevant sectors: steel, cement, chemicals, minerals and engineering. EPOS's main objective is to enable cross-sectorial Industrial Symbiosis (IS) and provide a wide range of technological and organisational options for making business and operations more efficient, more cost-effective, more competitive and more sustainable across process sectors. www.spire2030.eu/epos

COPRO (H2020 SPIRE-02-2016): Improved energy and resource efficiency by better coordination of production in the process industries (just started) – *INEOS partner*. CoPro will provide tools for the plant-wide optimisation of continuous and discrete decisions, will develop technology for balancing production and consumption in industrial parks for industrial symbiosis and will address power plant scheduling and demand side response. www.spire2030.eu/CoPro

STYLE (H2020 SPIRE-04-2014): Sustainability Toolkit for easY Life-cycle Evaluation. This project, *coordinated by BRITEST*, was a 2-year CSA bringing together industry partners from across the SPIRE sectors to identify, evaluate and propose a suitable collection of sustainability evaluation tools to meet the needs of industry project teams. Included in STYLE's recommendations were ways in which sustainability assessment could be integrated into the early-stages of SPIRE projects, rather than

focusing on detailed quantitative assessments towards the end of funded projects. STYLE also worked closely with the SAMT and MEASURE projects to develop joint recommendations, and hence can bring learning from these projects to SPRING too. www.spire2030.eu/style

DREAM – Design for Resource and Energy efficiency in cerAMic kilns. Funded under the SPIRE-04-2016 call, this project is led by SACMI, world leader in ceramics manufacturing equipment design and production, and has *CRIT as dissemination and exploitation leader*. Despite being fully targeted to ceramics technologies (and thus missing the trans-sectorial aspect of several SPIRE funded actions), the project contains also a dedicated market replication task, aiming to explore and pursue exploitation pathways in other process industries for selected key results. Thus, DREAM will provide a test bench for systematic trans-sectorial technology transfer, starting from a focused research activity. www.spire2030.eu/dream

1.3.4 – COORDINATION AND MANAGEMENT

Coordination and management of the project will be supplied by Britest. The organisation has, since its foundation in 2001, led collaborative projects and activities amongst its members, and has developed capabilities in the design and delivery of complex inter-disciplinary projects at the pre-competitive level.

Britest's membership includes a wide range of international industrial organisations both large and small (including Pfizer, AstraZeneca, Johnson Matthey, Fujifilm, Biogen, Victrex, Hovione, Stirling); international research and academic organisations (including ICES, Leeds, Nottingham and Limerick Universities). Through working with these organisations Britest has developed a distinctive knowledge and understanding for driving effective collaboration across industries and academic sectors. Britest brings this experience to project SPRING and will involve its member organisations as part of the wider stakeholder engagement activities.

Britest also has direct experience of CSA project coordination through the SPIRE-04-2014 project, STYLE.

A compact management structure for project SPRING has been chosen to reflect the fact that this is a CSA call with a limited number of core project partners, but with a larger stakeholder community interface. The management structure therefore consists of a General Assembly (GA), Industry Advisory Board (IAB) and a Stakeholder Forum (SF) – see section 3.4.

1.3.5 – GENDER

We envisage an excellent opportunity to explore the influence or not of gender aspects in impact delivery in RIA and IA collaborative projects. As such we will analyse the female participation in the existing SPIRE projects that have been funded to date and the different levels of involvement of females in the consortia (project management roles and other leadership roles, research, IPR, communication, dissemination, etc.), in addition to engagement with female stakeholders (industry leaders, etc.). We will also review the level of gender analysis activities being undertaken in this project at the implementation stage. Our aim will be to establish whether or not gender has an influence on certain aspects of success or not. With an increasing representation of females in leadership and decision making roles in the Process Industries, the ability to effectively address gender analyses in a meaningful way could have an influence on the ability to communicate and deliver impact.

Moreover, the SPRING consortium has a very good gender balance (45% of participants are female) and females occupy a number of key leadership roles in the consortium: Project Coordinator, Industry Advisory Board Chair, WP and task leaders.

2. IMPACT

2.1 EXPECTED IMPACTS

2.1.1 – CONTRIBUTION TO THE EXPECTED IMPACTS

Project SPRING will contribute to the expected impacts of the work programme in the following ways:

- *Speeding up industrial exploitation and take up of results of SPIRE PPP projects and facilitate cross-sectorial technology transfer.*

  Project SPRING will speed up industrial exploitation by developing guidance for SPIRE project participants on the classification of exploitable project outcomes, which together with an improved structure for capturing and communicating these on the spire2030.eu web portal, will enable decision makers from industry to more quickly and clearly identify outputs of potential benefit. Thematic workshops will bring together common threads from across the portfolio of SPIRE projects to facilitate cross-sectorial learning and technology transfer; specifically, with a trial targeted at finding outputs of relevance to the ceramics industry.

- *Stimulation of networks and alliances for further RTD and industrial innovation in the addressed technology and application areas.*

  Project SPRING will facilitate the development of new alliances and networks through the establishment of new SPIRE-level Network Groups (Coordinators, Sustainability Practitioners and SMEs) and through the thematic workshops, which aim to bring together projects from different funding calls but with common themes.

- *Added value beyond the original scope of the SPIRE PPP projects by exploiting synergies and sharing best practice, including on innovative business models. Increased public presence and awareness of SPIRE PPP activities.*

  Project SPRING will bring added value to projects through the development and publication of best practice guidance (on metrics, communicating impact, addressing industry hurdles etc.) and providing mechanisms for exploiting synergies between projects (e.g. the Coordinator Network Group). Awareness will be increased through improvements to the spire2030.eu web portal, making it easier for stakeholders to find outputs of interest, and through the thematic workshops and associated publicity.

- *More effective execution of activities of common interest, such as training & education, IPR management and standardisation.*

  Project SPRING will make it easier for projects to collaborate on areas of common interest, such as IPR management, data management and standardisation, through the SPIRE Coordinators Network Group. Improvements to the spire2030.eu web portal will enable project output material suitable for education and training to be structured and searchable for academia, industry and training providers. The guidelines developed in Project SPRING will support SPIRE project participants in developing education, training and other material related to exploitation and impact in a common format.

- *Anticipation of business trends and market prospects. Early awareness of key innovation developments.*

Project SPRING will review best practice for technology scanning and business trends, and test on case study projects, culminating in thematic workshops which will inform stakeholders of project outputs alongside innovation developments external to SPIRE. The learning from this will be incorporated into the guidance for SPIRE project participants to enable them to conduct similar horizon scanning activities.

- *Dissemination of project results beyond traditional dissemination models and timeframe of the projects.*

  Project SPRING will develop a framework for structuring project outputs for educational and training purposes on the spire2030.eu portal beyond the standard timeframe of the projects. This will build on the work of Educate to Innovate, which trialled approaches with the FP7 project 'F3 Factory'. The thematic workshop structure will also place a priority on best practice to engage with industry decision makers, methods for interaction and stepping away from 'project seminar' approaches. Social channels, such as LinkedIn groups, will also be utilised within SPRING to provide dissemination and input channels to stakeholder communities.

- *Development of training and innovation skills capacity in the process industry.*

  Project SPRING will develop guidance and frameworks that will enable training material from SPIRE projects to be more readily accessible through the spire2030.eu website. In addition, it will produce its own set of training resources to upskill those involved in SPIRE projects to enhance impact within and beyond the timescales of their project (e.g. Communicating Impact, Integrating Sustainability into Project Management, Business Opportunity Planning etc.)

2.1.2 – BARRIERS AND FRAMEWORK CONDITIONS

Project SPRING is directly addressing barriers to industry uptake of project outputs. SPRING will provide a structured means to facilitate ease of access to a broad portfolio of SPIRE project outcomes. Ultimately, the biggest barrier is industry wanting to help itself, i.e.

- Willingness of SPIRE project partners to apply the guidelines developed in SPRING to develop materials for enhanced impact
- Willingness of SPIRE project partners to undertake SPRING training
- Willingness of SPIRE project partners and the broader SPIRE process industry community to participate in Network Groups

By incorporating into the project an open Stakeholder Forum and Industry Advisory Board, SPRING consortium partners will be made aware, at inception and throughout the project, of many of the issues, concerns and uncertainties which could impair effective roll-out of the findings. The consortium will therefore be able to take appropriate action at an early stage to lower the identified barrier. The inclusion of A.SPIRE as a project partner also helps to ensure that the outputs of SPRING are complementary to the advice being given and requests for information and participation already being directed at SPIRE projects.

2.2 MEASURES TO MAXIMISE IMPACT

2.2.1. DISSEMINATION AND EXPLOITATION OF RESULTS

An important element of SPRING is ensuring that its impacts will endure beyond the timescale of the project, for many years to come.

A dedicated project website will be the principal means by which stakeholder engagement is encouraged and supported throughout the project and from which the key project outputs will be disseminated and exploited. The project website will be hosted though www.spire2030.eu to provide consistency with existing projects and to ensure maintenance and accessibility once the project has completed.

The SPIRE portal (www.spire2030.eu) has recently been upgraded to allow provision for interaction and communication between SPIRE stakeholders. Project SPRING will enable A.SPIRE to develop features and functionalities of the website portal consistent with best practice and to focus efforts on to elements that bring *added value* for the SPIRE community. For example, some networking groups may be better hosted through LinkedIn, to minimise the number of platforms that stakeholders use on a regular basis and to encourage broader participation beyond those already engaged in SPIRE activities; whereas, closer active collaboration between live SPIRE projects may be better done through the SPIRE portal to help manage confidential information and to link to other features of the portal, e.g. event calendars.

To encourage broad participation at stakeholder and industry advisory board workshops and meetings, provision has been made in the project budget to fund reimbursement of travel costs for non-consortium attendees, including SME's. A policy for this will be established by the project consortium at the start of the project.

Opportunities for remote participation of stakeholders via survey, website discussion forums, webinars etc., will also be established to ensure all potential stakeholder groups/ individuals can both contribute to – and benefit from – results emerging from the project. Non-traditional dissemination channels through videos, blogs and social media will be utilised to make project information and outputs as engaging as possible.

The consortium will also seek opportunities for preparation of papers suitable for publication, as well as for case studies, to raise awareness and promote uptake of enhanced project impact guidelines.

Table 3 – SPRING Primary Exploitable Results

Exploitable Results	Measure	Owners
Guidance on how to measure progress/ impact/ success of projects	Best Practice Manual (report)	Open Access
High level framework for getting targeted awareness of SPIRE projects' outputs	Interactive website feature; Promotional video targeted at industry	A.SPIRE; Open Access
Framework for extracting training/ teaching resources from SPIRE projects	Interactive website feature; Promotional video targeted at academics	A.SPIRE; Open Access
Model for mapping project outputs against industry needs	Thematic workshops; report	Open Access

Guidance for understanding business barriers to uptake	Best Practice Manual	Open Access
Training workshop and presentation package for communicating value and impact	Training Package available as a SPIRE resource	A.SPIRE; Open Access
SPIRE SME Working Group	Member list	A.SPIRE
SPIRE Coordinators Network Group	Member list	A.SPIRE
SPIRE Sustainability Practitioners Working Group	Member list	A.SPIRE
Strategies for addressing policy gaps and future SPIRE needs	Report	Open Access

The SPRING exploitable outputs also have the potential to contribute more broadly to the wider Horizon2020 objectives of driving economic growth, creating jobs and removing barriers to innovation to make it easier for the public and private sectors to work together in delivering innovation. To this end, not only will SPRING outputs be thoroughly trialled within the 2-year timeframe of the project but steps will also be taken to encourage their exploitation beyond the timeframe of SPRING for all future SPIRE projects and further, where applicable, to other H2020/ FP9 future projects.

Realisation of the full impact of project SPRING will still be dependent on the willingness of SPIRE stakeholders to use the guidance, training, website frameworks, etc. to enhance their own projects. However, A.SPIRE will continue to play an important role in the exploitation of SPRING results once the project has completed, taking ownership of the resources to help SPIRE projects, ensuring longevity of SPRING's legacy.

Data Management

The majority of the data collected through SPRING will relate to the outputs from other SPIRE projects. SPRING will be working to ensure that relevant data from projects will be preserved through the spire2030.eu portal.

The SPRING project will participate in the Open Research Data Pilot initiative. Effective exploitation of data depends upon proper data management and curation and as such as Data Management Plan will be drawn up and managed through *Task 1.3: Data and Knowledge Management*. A formal **data plan** will be drawn up and implemented during the project that will include: which data will be generated during the research; metadata, standards and quality assurance measures; plans for sharing data; copyright and intellectual property rights of data; data storage and back-up measures; data management roles and responsibilities; costing or resources needed. Indeed, data management will be embedded in the wider knowledge management of the project, and it will follow the Guidelines on Open Access to Scientific Publications and Research data in Horizon H2020. The ethical standard and guidelines of Horizon2020 will be rigorously applied, regardless of the country in which the research is carried out.

Knowledge Management and Protection

As a CSA, most of the knowledge generated within SPRING will be in an Open Access format to ensure maximum benefit to the broad SPIRE community. SPRING will work with individual SPIRE case study projects to

establish which outputs are available for open dissemination. However, that said, through *Task 1.3: Data and Knowledge Management*, under the leadership of the Project Coordinator, SPRING will implement a strategy for knowledge management in order to ensure that no confidential or restricted output material from test/case study projects is put into the public domain, but will look for opportunities to translate and genericise outputs to make them suitable for wider dissemination and exploitation.

In relation to IPR rules, we will follow the Horizon 2020 default rules from the model grant agreement.

As such, in relation to background IPR and access rights, whereby **'Background'** means any data, know-how or information-whatever its form or nature (tangible or intangible), including any rights such as intellectual property rights –that is a) held by the beneficiaries before they acceded to the Agreement, and b) needed to implement the action or exploit the results relation to access rights to Background, the consortium will comply with the H2020 default rules, whereby on the one hand, the partners/beneficiaries will provide each other access — on a royalty-free basis — to Background needed to implement their own tasks under the action/project. On the other hand, the partners will give each other access — under fair and reasonable conditions — to background needed for exploiting their own results. The consortium will declare an up to date status of Background before the Grant Agreement (GA) is signed, and at proposal stage it has been confirmed that all partners have the access they need to each other's background.

Results definition, protection, ownership and access rights: '**Results**' means any (tangible or intangible) output of the action such as data, knowledge or information — whatever its form or nature, whether it can be protected or not — that is generated in the action/project, as well as any rights attached to it, including intellectual property rights. As discussed previously, the Results being outputted from the project will be put into the public domain ensuring that no confidential information from the project is revealed. To this end, to ensure that the Results do not trespass into the owned knowledge space of a third party, an internal agreement to the produced outputs will be sought upon the participating entities. In case results that can be exploitable (methodologies, strategies) appear during the course of the project, those **results shall be owned by the participant generating them**. To avoid doubt or confusion over ownership, a protocol of procedures will be defined that each partner will follow to clearly show the development of the generation of knowledge or Results. In cases where a Result is generated jointly (i.e. where the separate parts of some result cannot be attributed to different participants), it will be **jointly owned**, with the allocation and the terms of exercising this joint ownership to be included in a Joint Ownership Agreement that will be drawn up.

Open access to peer-reviewed publications: Funding is available through Leeds University Business School to cover required open access article processing charges. Additionally, the University of Leeds is a partner in the White Rose repository, which is categorised as a green standard repository with respect to open access.

2.2.2. COMMUNICATION ACTIVITIES

In recognition of the SPRING project structure, whereby the SPRING outputs are being generated to help direct the existing activities of A.SPIRE, communication activities will be managed between WP1 (Coordination and Management), WP4 (Output Generation) and WP5 (Stakeholder Engagement). BRITEST will manage intraconsortium communication within WP1 tasks and provide the link to their leadership of WP4, where outputs and publicity material will be generated for external communication. WP5, led by A.SPIRE, will act as the route to gather input from stakeholders and be the outlet for dissemination and exploitation of the outputs and publicity.

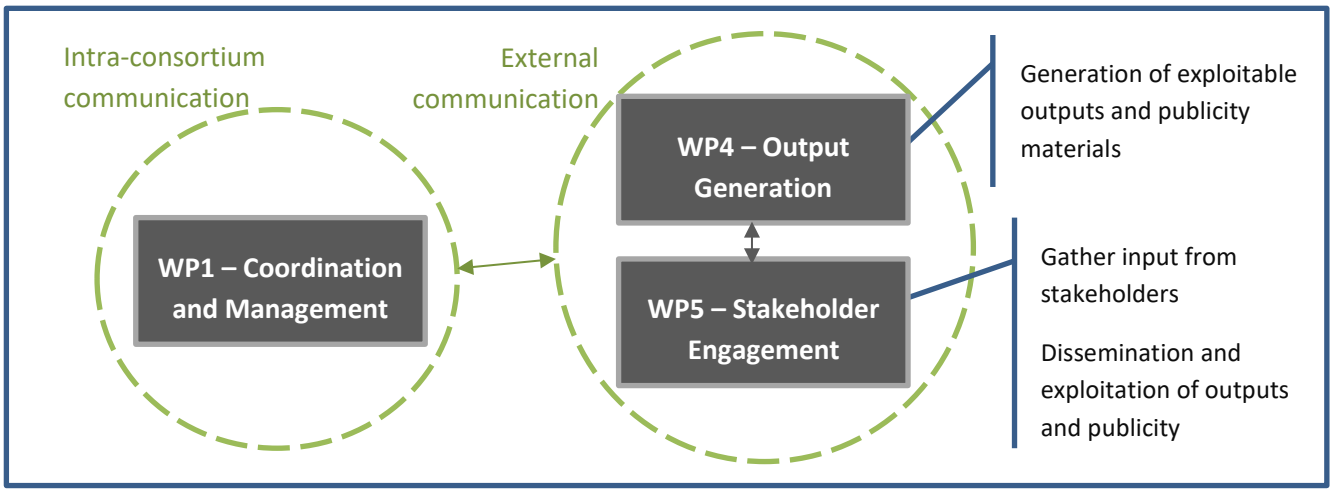


Figure 4 – Project SPRING Communication Structure

The programme of communication activities will ensure that the project objectives, progress and outputs are disseminated openly and professionally to a broad stakeholder community from project inception and throughout the project to ensure lasting impact beyond its completion.

Intra-consortium information and communication sharing

A secure project 'portal' will be established by BRITEST to facilitate the effective communication and distribution of information, reporting and data storage to and amongst project partners. This will act as the principal repository for project documentation and provide a platform for work package leaders to communicate and coordinate information. It will also be used to facilitate dialogue amongst researchers and case study project partners in the project and information exchange across work packages.

External information and communication sharing

An open access project website will be established through spire2030.eu to raise awareness of the project and facilitate the professional and effective communication of project objectives, progress and outputs. The website will be the principal means by which non-confidential project information is communicated externally and for which stakeholder engagement is promoted and maintained. Additional dissemination of project deliverables will be done via symposia, workshops and conferences where applicable.

A.SPIRE, as WP5 leaders, will use their existing communication channels with the SPIRE community to aid engagement with stakeholders and dissemination of project outputs. This includes the SPIRE newsletter, PPP information days, SPIRE project workshops, partnering events and General Assembly meetings.

In addition to the project website and SPIRE channels, dissemination will be actively sought via stakeholders and European/ National Technology Platforms, e.g. SusChem using their respective website, social and other communication media.

Communication links with other Horizon 2020 Projects

A.SPIRE will act as the main channel to other European FP7 and Horizon 2020 projects where there are opportunities for exchange of knowledge and approaches. CRIT will act as a link to H2020 Factories of the Future projects (through their 'FoF-IMPACT' project), bringing insight from a similar *impact enhancement* call in the FoF sectors. LEEDS will act as a link to broader European funded projects where impact enhancement methods have been evaluated.

Comprehensive communication activities will ensure that we effectively promote our project and its outputs, as well as its benefits and impacts to the stakeholders that we want to reach and secure engagement with. At this stage, we have mapped out a preliminary plan of the communication activities (*Table 2.2.2 – Outline Plan of the SPRING Communication Activities*) that will be performed for promoting the project and its findings during the period of the grant, based on a process of analysing: **Why** – the purpose of the communication action; **What** will be communicated – the message; **To whom** – the audience; **How** – the method; **When** – the timing. As part of the workplan and our communication structure, we will build on, implement and report on this plan.

Table 2.2.2 – Outline Plan of the SPRING Communication Activities

How (Method)	Why (Purpose)	To whom (Audience)	What (Message)	When	Metrics
Project Website	Awareness Inform Engage Promote	SPIRE projects; Other Relevant H2020 Projects; A. SPIRE members; Wider Process Industries; Other PPP, Associations and Multipliers; Policy makers.	General project information, objectives, impacts, consortium, access to public deliverables and outputs, project news and events, engagement information	First version launched by M2. Ongoing updating and throughout the project	Checking usage logs & no. of project consultations
Video/web viral	Awareness Inform Promote		Project results, impact for SPIRE, benefits for industry, academics, environment, public	Expected by M23 (final date will be agreed during the project). Available from website & other digital platforms	No. of viewings
Newsletters	Awareness		Project progress and updates, Industry testimonials and interviews, Upcoming Events, etc.	Every 6 months (disseminated through SPIRE, partners' contacts; downloadable from web, etc.)	No. of consultations
Press Releases	Awareness	General public	Public interest content about the impacts of the project for sustainability, competitiveness, growth, jobs, benefits of EU funded R&D.	At KO, and in line with public interest developments in the project, events, etc.	No. of hits on web
Project Blog	Awareness Inform Engage	All External Stakeholders	Posts about developments, updates, observations, conclusions	At least one post per quarter (members of the consortium team will write different blog articles)	Readership & No. of consultations
Programme and Network meetings	Engage	SPIRE and other H2020/National Projects; EC stakeholders; Policy; Industry networks.	Best Practice in Impact Enhancement in Research and Innovation Actions and Innovation actions	At least 3 over course of project: e.g. Meetings of A.SPIRE; Commission Impact Workshops etc.	No. of engagements
Conference Posters & Presentations	Engage Promote	Research community	Project methodology, results and conclusions. Note: presentations will also be disseminated on Slideshare that has an average of 60 million unique visitors per month.	e.g. Industrial Technologies (Biannual EU Conference on industrial R&D), Tecnargilla (Biannual fair on ceramics technologies), ACHEMA etc.	No. of engagements and No. of hits on web
Peer reviewed papers	Inform			e.g. International Journal of Innovation Management, Journal of Innovation & Knowledge etc.	No. of citations
Articles in Magazines	Inform	SPIRE projects; Other Relevant H2020 Projects; A. SPIRE members; Wider Process Industries; Other PPP, Associations and Multipliers; Policy makers.	Best Practice in Impact Enhancement in Research and Innovation Actions and Innovation actions. Knowledge/Lessons learned in the project.	e.g. Research*eu, sector specific publications, SusChem online articles etc.	No. hits on web and consultations
Workshop and Training sessions	Engage Inform			At least 3 over course of the project to coincide with key outputs and milestones	No. of attendees & No. of engagements
Case studies/ Insight Summaries	Inform		Snapshot of project findings, with reference to case studies.	Coincide with guidance development	No. web downloads & consultations
Social Media/ Digital campaign	Awareness Inform Engage Promote		Tailored digital campaigns in social media channels using infographics, storytelling, cast studies, etc. to communicate vision, ambition, benefits and impacts of our project, etc.	A series of campaigns will be scheduled to coincide with key milestones, activities and events	No. of likes, comments, etc.

Reports and other project documents	Inform	Available to all external stakeholders	Public deliverables (from the deliverables list)	Upon completion, according to the deliverables list	No. web downloads & consultations

3. IMPLEMENTATION

3.1 WORK PLAN – WORK PACKAGES AND DELIVERABLES

The table below summarises the work packages defined by the Project SPRING consortium. In overview only, the packages are:

WP1 – Coordination and Management	Covering management of the project, its activities, its resources, its internal communications and its links to the Commission	BRITEST lead
WP2 - Research	Gathering knowledge from partners within the consortium, the SPIRE and FoF projects that they represent, stakeholders in the SPIRE community and broader project and policy research to propose best practice methods	LEEDS lead
WP3 – Testing and Validation	Testing recommended methods and guidance on case study SPIRE projects	IRIS lead
WP4 – Output Generation	Developing the exploitable outputs for SPRING. These include guidance for current and future SPIRE projects, frameworks for the spire2030.eu website and training	BRITEST lead
WP5 – Stakeholder Engagement	Stimulating broad engagement throughout the project, providing a route to gather input from the broader SPIRE community and for outputs from the project to get back to the stakeholders	ASPIRE lead

The structure for SPRING is a matrix of Work Packages, through which the project outputs are produced:

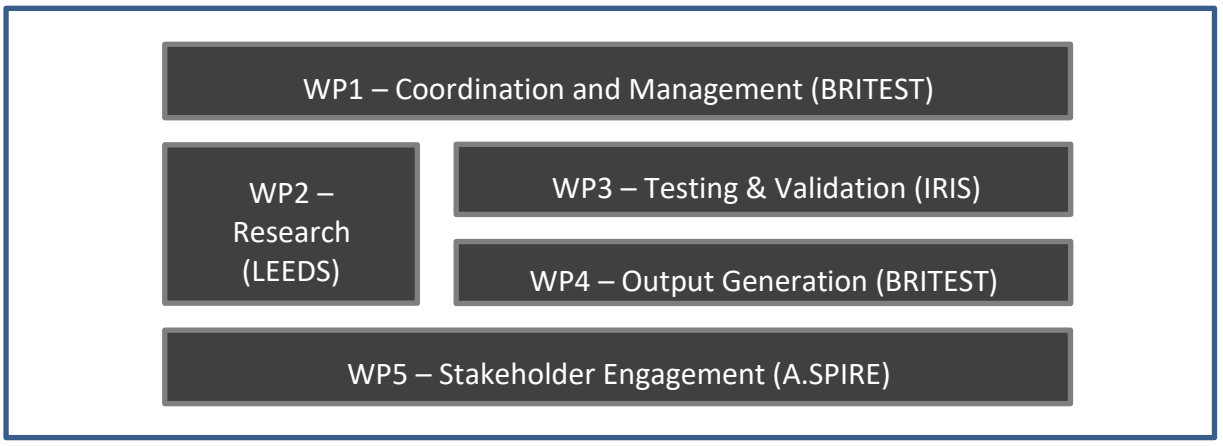
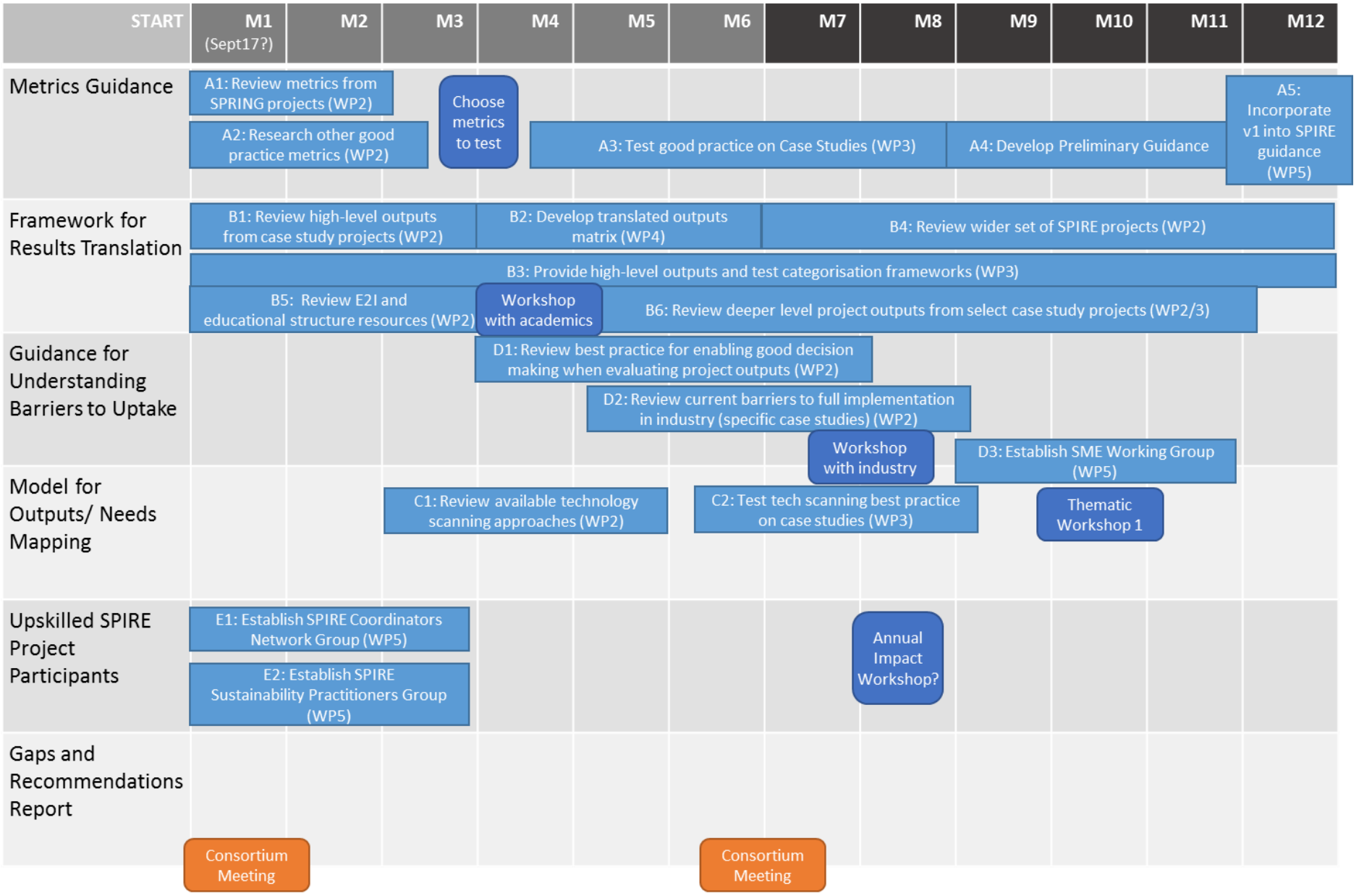
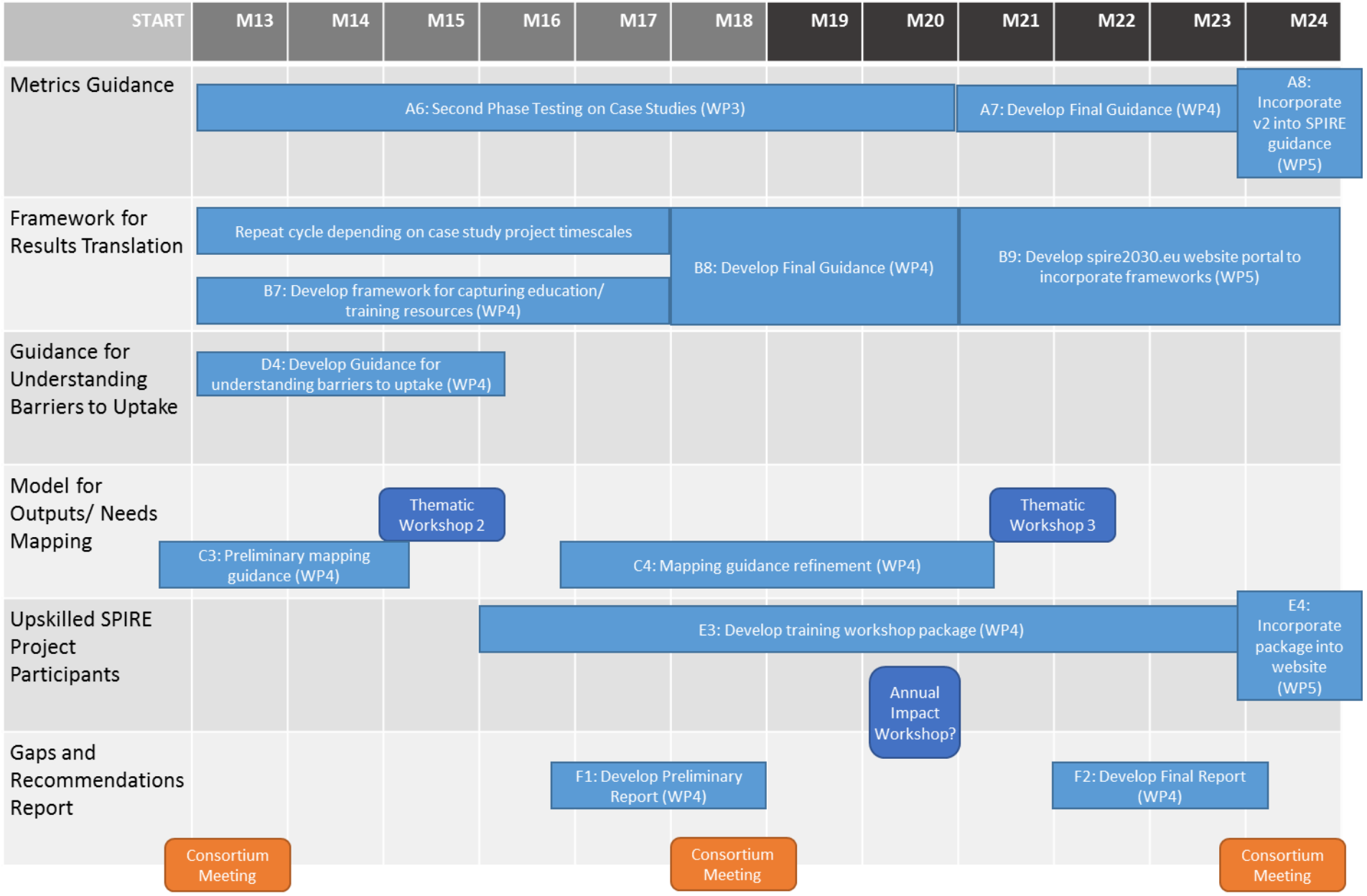


Table 5 - Work Package Structure (& lead partner)

PROJECT SPRING – HIGH LEVEL TIMELINE – YEAR 1


START
M1 (Sept17?)
M2
M3
M4
M5
M6
M7
M8
M9
M10
M11
M12
Metrics Guidance
A1: Review metrics from SPRING projects (WP2)
A2: Research other good practice metrics (WP2)
Choose metrics to test
A3: Test good practice on Case Studies (WP3)
A4: Develop Preliminary Guidance
A5: Incorporate v1 into SPIRE guidance (WP5)
Framework for Results Translation
B1: Review high-level outputs from case study projects (WP2)
B2: Develop translated outputs matrix (WP4)
B4: Review wider set of SPIRE projects (WP2)
B3: Provide high-level outputs and test categorisation frameworks (WP3)
B5: Review E2I and educational structure resources (WP2)
Workshop with academics
B6: Review deeper level project outputs from select case study projects (WP2/3)
Guidance for Understanding Barriers to Uptake
D1: Review best practice for enabling good decision making when evaluating project outputs (WP2)
D2: Review current barriers to full implementation in industry (specific case studies) (WP2)
Workshop with industry
D3: Establish SME Working Group (WP5)
Model for Outputs/ Needs Mapping
C1: Review available technology scanning approaches (WP2)
C2: Test tech scanning best practice on case studies (WP3)
Thematic Workshop 1
Upskilled SPIRE Project Participants
E1: Establish SPIRE Coordinators Network Group (WP5)
E2: Establish SPIRE Sustainability Practitioners Group (WP5)
Annual Impact Workshop?
Gaps and Recommendations Report
Consortium Meeting
Consortium Meeting

PROJECT SPRING – HIG H LEVEL TIMELINE – YEAR 2

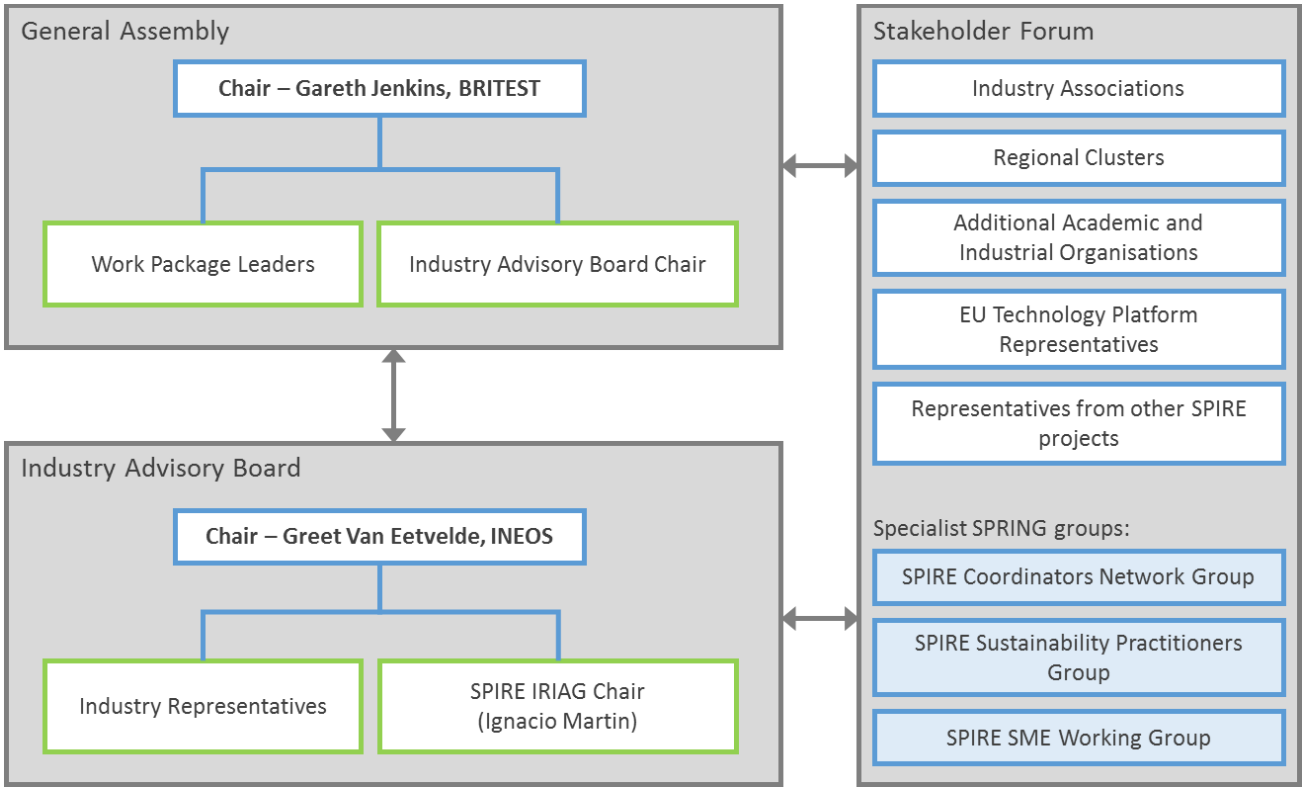


3.2 MANAGEMENT STRUCTURE AND PROCEDURES

Project SPRING's management structure will be a General Assembly (GA), Industry Advisory Board (IAB) and Stakeholder Forum (SF).

It is anticipated that the General Assembly and Industry Advisory Board will meet at the inception of the project and at six monthly intervals thereafter. IAB meetings may also be timed to coincide with general SPIRE meetings, to ensure high attendance and efficient use of industrialist's time.

A compact management structure for project SPRING has been chosen to reflect the fact that this is a CSA call with a limited number of core project partners but with a large stakeholder community interface.



The **General Assembly** is the ultimate decision-making body of the Consortium. Each project partner (legal entity) appoints one representative to be a member of the GA. It is anticipated that the GA will be chaired by a senior employee of the organisation carrying out WP1 (BRITEST) – to be confirmed by a vote at the first meeting. The GA will be the key body for significant decisions concerning the whole project and for the strategic orientation of the project. Examples include changes to the consortium plan, decisions about the entry of a new party to the consortium, the declaration of a party as defaulting party, the withdrawal of a party from the consortium or the termination of the project. Under exceptional circumstances, preliminary decisions can be made via a remote meeting of the General Assembly members. The following operational principles are proposed for SPRING. At the project inception, these will be examined in detail, modified as necessary and ratified by the full General Assembly:

- Each member has one vote
- Where possible, General Assembly members will attempt to reach a conclusion by discussion and agreement. If agreement cannot be reached, all members will vote.

- Decisions will be reached by a simple majority, with the chair having a casting vote; major project changes will require the vote to be unanimous.
- In the event of a dispute, those Consortium members who are not directly involved in the dispute will act as an Arbitration Panel to resolve the dispute.

The **Industry Advisory Board** will consist of a minimum of three industry representatives from different SPIRE sectors, along with a representative from SPIRE's IRIAG (Industry Research and Innovation Advisory Group) – Ignacio Martin, IRIAG Chair. The SPRING Industry Advisory Board will be chaired by Greet Van Eetvelde (INEOS) and will be tasked with ensuring that SPRING's work is focused on the *barriers to industrial uptake* of project findings. The IAB will also be used to help broaden and strengthen links to other industry stakeholders. BRITEST will act as secretariat to the IAB.

BRITEST, as **Project Coordinator**, will provide inputs to the Commission.

Work Package Leaders (WPL) will be responsible for coordinating, monitoring and reporting of the Work Package. Tasks of the Work Package Leaders are the implementations of the Work Programme and the reporting on deliverables and milestones to the Project Coordinator.

The **Stakeholder Forum (SF)** plays an important role in linking the SPRING project to organisations, individuals and activities beyond the project consortium. The Stakeholder Forum will act as a two-way communication channel to both gather the thoughts and experiences of the members, and to inform the wider stakeholder community as to the findings from SPRING project. The forum will be based as an online group, but specialist sub-groups will be assembled from the forum to tackle specific tasks in SPRING, e.g. academic course developers to consult on the framework for project outputs.

Three specific specialist stakeholder groups will be established as part of SPRING that should continue beyond the timeframe of SPRING:

- **SPIRE Coordinators Network Group:** This group should enable more cross-project collaborations and provide a route for gathering best practice and recommendations dissemination.
- **SPIRE Sustainability Practitioners Group:** This group should help bring together those tasked with sustainability evaluations in SPIRE projects, aiding the uptake of SPIRE-04-2014 projects' recommendations and encouraging best practice sharing and more consistent approaches to integrating sustainability into early project developments.
- **SPIRE SME Working Group:** This group should help gather views from the SME members of SPIRE as to specific challenges for smaller organisations getting project outputs into industry.

3.3 CONSORTIUM AS A WHOLE

BRITEST is ideally placed to coordinate this CSA. BRITEST started life as an Open Innovation collaboration, working with industry and academia on a pre-competitive basis, allowing it to build a role as a 'trusted intermediary'. BRITEST is a not-for-profit organisation, with a core aim of empowering industry and academia to translate technical innovations and issues into clear structured knowledge that is understandable and accessible by cross-disciplinary teams. BRITEST strives to get project teams delivering better projects; not for them to need consultants on a regular basis.

BRITEST represents its own consortium of members and associates from the process industries and academia (e.g. Pfizer, Johnson Matthey, FujiFilm, Astra Zeneca etc.), providing further opportunities to get industry input to SPRING. A large part of BRITEST's work is coordinating projects between BRITEST members and with external consortia. Consequently, BRITEST see this project as an opportunity to raise the game, to understand and then develop best practice for getting collaborative project results back into industry, delivering real value.

The SPRING consortium represents a broad selection of existing SPIRE projects (including STYLE, EPOS, IbD, ProPAT, SHAREBOX, and DREAM), keen to come together to look for further collaboration and exploitation opportunities.

Large industry is represented through INEOS, BRITEST Members and the Industry Advisory Board; SMEs are represented through IRIS, CRIT and the SME Working Group; academia is represented through LEEDS and academic stakeholders. A.SPIRE provide a link to all other SPIRE PPP stakeholders.

IRIS is a good example of a technology SME which is actively leading and implementing three SPIRE projects, in order to showcase the relevant role and opportunities that are open to SMEs within the SPIRE PPP. Technology SMEs often hold the Key Enabling Technologies (KETs) to resolve the challenges faced by the European Process Industries and can be very flexible and innovative allies to Large Enterprises in terms of technological innovations.

CRIT and IRIS both have strong links to the ceramics sector, which has been one of the least represented sectors in SPIRE projects to date, hence SPRING will provide an opportunity to make a specific trial of identifying and/ or translating outputs from other SPIRE sectors relevant to ceramics.

All partners are members of SPIRE, so have interests in helping the PPP achieve its Roadmap goals.

The focus of the SPRING project is *what is needed to get projects outputs adopted by industry*. This is reflected in the industry-biased consortium. In addition, the Leeds University contribution is led by their Business School to keep the focus on business exploitation of project outputs.